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System and method for performing tissue treatment using powered treatment devices

Abstract

A powered treatment device for restructuring and revitalizing fascia tissue. The powered treatment device may include a housing, an actuator, and a tissue treatment assembly. The actuator may be configured to rotate and may be coupled to and disposed substantially within the housing. The tissue treatment assembly may include a panel and a plurality of finger members. The plurality of finger member may be fixedly coupled to the panel at a proximal end of the respective finger members. The finger members may be rigid and extend away from the panel. Each one of the finger members may have a respective central axis that extends from the proximal end to a distal end of the respective finger member. At least one of the central axes may extend at least partly along a radial direction and/or a circumferential direction of the panel.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
D21551	12/1891	Young	N/A	N/A
687363	12/1900	Wirt	N/A	N/A
912016	12/1908	Miller	N/A	N/A
965564	12/1909	Coates	N/A	N/A
1002509	12/1910	Fitz	4/258	A61J 19/00
1018939	12/1911	Sterrick	N/A	N/A
D55452	12/1919	Chase	N/A	N/A
1523979	12/1924	Ryan	N/A	N/A
1713756	12/1928	Hassler	N/A	N/A
2168975	12/1938	Dumont	N/A	N/A
D140384	12/1944	Latshaw	N/A	N/A
2468327	12/1948	Hartung	N/A	N/A
2480023	12/1948	Holden	15/159.1	A61H 7/002
2972347	12/1960	McNair	N/A	N/A
D197889	12/1963	Hass	N/A	N/A
D208894	12/1966	Wedermeyer	N/A	N/A
D213690	12/1968	Eckstrom	N/A	N/A
D233632	12/1973	Kirk	N/A	N/A
3856002	12/1973	Matsumoto	N/A	N/A
3862662	12/1974	Kern	N/A	N/A
D234724	12/1974	Tavel	N/A	N/A

3906940	12/1974	Kawada	604/315	A61H 9/005
4023567	12/1976	Wessel	N/A	N/A
4091805	12/1977	Clark	N/A	N/A
4173217	12/1978	Johnston	N/A	N/A
D260928	12/1980	Gueret	N/A	N/A
4345757	12/1981	Lo Voi	N/A	N/A
D269375	12/1982	Masuda	N/A	N/A
D270279	12/1982	Hamilton	N/A	N/A
D272090	12/1983	Hosid	N/A	N/A
4454867	12/1983	Swanson	N/A	N/A
4493315	12/1984	Iwahashi	N/A	N/A
D279126	12/1984	Kaeser	N/A	N/A
D280434	12/1984	McManaway	N/A	N/A
D280664	12/1984	Ishida	N/A	N/A
D281815	12/1984	Conn et al.	N/A	N/A
D282007	12/1984	McKinney	N/A	N/A
D300853	12/1988	Smal	N/A	N/A
D315408	12/1990	Duk	N/A	N/A
D319645	12/1990	Geib	N/A	N/A
5044356	12/1990	Fishman et al.	N/A	N/A
5103809	12/1991	DeLuca et al.	N/A	N/A
D331466	12/1991	Doria	N/A	N/A
D333351	12/1992	Tsou	N/A	N/A
D342319	12/1992	Cheng	N/A	N/A
5285774	12/1993	Stachurski	N/A	N/A
D345054	12/1993	Spence	N/A	N/A
D346866	12/1993	Lotuaco	N/A	N/A
D349576	12/1993	Heiligenstein et al.	N/A	N/A
D349771	12/1993	Heiligenstein et al.	N/A	N/A
D350608	12/1993	Yu	N/A	N/A
D350851	12/1993	Spence	N/A	N/A
D351658	12/1993	Beckman	N/A	N/A
5352188	12/1993	Vitko	N/A	N/A
D359358	12/1994	Pecora et al.	N/A	N/A
5445596	12/1994	Grace	N/A	N/A
5483719	12/1995	Ikemoto et al.	N/A	N/A
D369447	12/1995	Mann	D32/19	N/A
D373197	12/1995	Schepper	N/A	N/A
D374084	12/1995	Chen	N/A	N/A
D376853	12/1995	Hsia	N/A	N/A
5588953	12/1995	Chang	N/A	N/A
5647841	12/1996	Groenewold et al.	N/A	N/A
D384157	12/1996	Kusnets et al.	N/A	N/A
5674261	12/1996	Smith	N/A	N/A
D387873	12/1996	Lee	N/A	N/A
D390966	12/1997	Huang	N/A	N/A
D391645	12/1997	Chien	N/A	N/A
D392424	12/1997	Burys	N/A	N/A

5728050	12/1997	Lin	N/A	N/A
5730708	12/1997	Spratt	N/A	N/A
D396296	12/1997	Breznik	N/A	N/A
D398401	12/1997	Antoskow	N/A	N/A
5823145	12/1997	Hingiss	N/A	N/A
D403429	12/1997	Blanchard	N/A	N/A
D403430	12/1997	Chien	N/A	N/A
D404495	12/1998	Chien	N/A	N/A
D408925	12/1998	Berry et al.	N/A	N/A
D419681	12/1999	Chen	N/A	N/A
D422084	12/1999	Mickelson	N/A	N/A
D422086	12/1999	Chen	N/A	N/A
D422365	12/1999	Chen	N/A	N/A
D427318	12/1999	Chen	N/A	N/A
D428997	12/1999	Berry et al.	N/A	N/A
D430936	12/1999	Noble	N/A	N/A
6139553	12/1999	Dotan	N/A	N/A
D435110	12/1999	Chen	N/A	N/A
D435295	12/1999	Yoo	N/A	N/A
6179229	12/2000	Desmond	N/A	N/A
6190339	12/2000	Imazaike et al.	N/A	N/A
D443067	12/2000	Chen	N/A	N/A
6241693	12/2000	Lambden	N/A	N/A
6241696	12/2000	York	N/A	N/A
D444883	12/2000	Chen	N/A	N/A
6267738	12/2000	Louis	N/A	N/A
D447809	12/2000	Gladieux et al.	N/A	N/A
D449695	12/2000	Ponton et al.	N/A	N/A
D454959	12/2001	Harris et al.	N/A	N/A
6357075	12/2001	Kaizuka	N/A	N/A
D455837	12/2001	Kim	N/A	N/A
6412997	12/2001	Berke et al.	N/A	N/A
D466612	12/2001	Harris et al.	N/A	N/A
D468084	12/2002	Lin	N/A	N/A
D468829	12/2002	Chen	N/A	N/A
D469879	12/2002	Chen	N/A	N/A
6536970	12/2002	Hauser et al.	N/A	N/A
D473657	12/2002	Lewis-Jonsson	N/A	N/A
6543954	12/2002	Owings	N/A	N/A
D477414	12/2002	Chen	N/A	N/A
6588964	12/2002	Au et al.	N/A	N/A
D479607	12/2002	Chen	N/A	N/A
D480148	12/2002	Chen	N/A	N/A
D484608	12/2002	Tinsley	N/A	N/A
D486236	12/2003	Nan	N/A	N/A
6758826	12/2003	Luettgen et al.	N/A	N/A
D495803	12/2003	Gough et al.	N/A	N/A
D499810	12/2003	Chien	N/A	N/A
D504954	12/2004	Park	N/A	N/A
D510440	12/2004	Huang	N/A	N/A

6974427	12/2004	Lapham	N/A	N/A
D515219	12/2005	Nan	N/A	N/A
D515702	12/2005	Nan	N/A	N/A
7003849	12/2005	Cohen et al.	N/A	N/A
7037016	12/2005	Nelson	N/A	N/A
D523962	12/2005	Huang	N/A	N/A
D530428	12/2005	Huang	N/A	N/A
D530822	12/2005	Inubushi	N/A	N/A
D532113	12/2005	Huang	N/A	N/A
7153282	12/2005	Dudley	N/A	N/A
D537892	12/2006	Moses	N/A	N/A
D538435	12/2006	Wang et al.	N/A	N/A
7238163	12/2006	Fried et al.	N/A	N/A
D548849	12/2006	Huang	N/A	N/A
7291101	12/2006	Deal	N/A	N/A
D556913	12/2006	Laituri	N/A	N/A
D558355	12/2006	Huang	N/A	N/A
D565304	12/2007	Stewart	N/A	N/A
D582049	12/2007	Ferber et al.	N/A	N/A
D590510	12/2008	Johansen et al.	N/A	N/A
D599061	12/2008	Smith	N/A	N/A
D600357	12/2008	Chaput	N/A	N/A
D613416	12/2009	Schupman	N/A	N/A
D626240	12/2009	Copeland et al.	N/A	N/A
D627897	12/2009	Yde et al.	N/A	N/A
D627898	12/2009	Aulwes et al.	N/A	N/A
D630761	12/2010	Marshall	N/A	N/A
D633217	12/2010	Robertson et al.	N/A	N/A
D643933	12/2010	Watabe	N/A	N/A
D648860	12/2010	Donahue et al.	N/A	N/A
D650488	12/2010	Donahue et al.	N/A	N/A
D652525	12/2011	Caggiano et al.	N/A	N/A
D656277	12/2011	Mckee	N/A	N/A
D664298	12/2011	Grabes et al.	N/A	N/A
D672925	12/2011	Hungerford	N/A	N/A
D673285	12/2011	Rivero	N/A	N/A
8337438	12/2011	Schupman	N/A	N/A
D673687	12/2012	Zeng et al.	N/A	N/A
D674107	12/2012	Prudent et al.	N/A	N/A
D674905	12/2012	Lowsky	N/A	N/A
D674911	12/2012	Caldwell	N/A	N/A
D676975	12/2012	Bickford	N/A	N/A
D677391	12/2012	Bickford	N/A	N/A
D678540	12/2012	Bickford	N/A	N/A
8419663	12/2012	Eitzen	N/A	N/A
D692569	12/2012	Samurin	N/A	N/A
D694415	12/2012	Viner et al.	N/A	N/A
D703825	12/2013	Barrett	N/A	N/A
D704852	12/2013	Yang	N/A	N/A
D708755	12/2013	Kim	N/A	N/A

D712053	12/2013	Matsushita	N/A	N/A
D712058	12/2013	Huang	N/A	N/A
8808208	12/2013	Mouatt	N/A	N/A
D716467	12/2013	Kanbar et al.	N/A	N/A
D719664	12/2013	Riftin et al.	N/A	N/A
D721420	12/2014	Hawkins	N/A	N/A
D721421	12/2014	Hawkins	N/A	N/A
D721440	12/2014	Pursel et al.	N/A	N/A
8954155	12/2014	Campbell	607/90	A61N 7/00
D723709	12/2014	Topolovac et al.	N/A	N/A
D724752	12/2014	Huang	N/A	N/A
D725788	12/2014	Matsushita	N/A	N/A
D732181	12/2014	Viner et al.	N/A	N/A
9050240	12/2014	Howsam	N/A	N/A
D735818	12/2014	Black	N/A	N/A
D739546	12/2014	Cheung	N/A	N/A
9125481	12/2014	Watanabe et al.	N/A	N/A
D742534	12/2014	Bains	N/A	N/A
D744663	12/2014	Holland	N/A	N/A
9237981	12/2015	Clementes	N/A	N/A
D750796	12/2015	Bains	N/A	N/A
D750841	12/2015	Lewis	N/A	N/A
D752237	12/2015	Cole et al.	N/A	N/A
D765872	12/2015	Blaskovich et al.	N/A	N/A
D773682	12/2015	Faussett	N/A	N/A
D776824	12/2016	Black	N/A	N/A
D777939	12/2016	Black	N/A	N/A
D785193	12/2016	Cole et al.	N/A	N/A
D790724	12/2016	Black	N/A	N/A
D805211	12/2016	Wu	N/A	N/A
9931003	12/2017	Sueyoshi	N/A	B01F 33/5011
D818588	12/2017	Black	N/A	N/A
D818600	12/2017	Black	N/A	N/A
D819825	12/2017	Black	N/A	N/A
D819826	12/2017	Black	N/A	N/A
D825929	12/2017	Julemont	N/A	N/A
D829445	12/2017	Kern	D4/127	N/A
D830568	12/2017	Black	N/A	N/A
D834832	12/2017	Kim	N/A	N/A
D841176	12/2018	Gaquiere et al.	N/A	N/A
D842489	12/2018	Spewock et al.	N/A	N/A
D843002	12/2018	Yarborough et al.	N/A	N/A
D844163	12/2018	Du	N/A	N/A
D844797	12/2018	Matsushita	N/A	N/A
D845002	12/2018	Disanti	N/A	N/A
D845499	12/2018	Wersland et al.	N/A	N/A
D845500	12/2018	Wersland et al.	N/A	N/A
D849260	12/2018	Wersland et al.	N/A	N/A
D849261	12/2018	Wersland et al.	N/A	N/A

D849263	12/2018	Soutelo Gomes	N/A	N/A
D850640	12/2018	Wersland et al.	N/A	N/A
D850641	12/2018	Black	N/A	N/A
10322057	12/2018	Black	N/A	N/A
D853576	12/2018	Ma	N/A	N/A
D855197	12/2018	Park	N/A	N/A
10383486	12/2018	Nichols	N/A	N/A
D858786	12/2018	Luo et al.	N/A	N/A
D858789	12/2018	Logan	N/A	N/A
D861182	12/2018	Wersland et al.	N/A	N/A
D863587	12/2018	Yee	N/A	N/A
D865203	12/2018	Eschbach et al.	N/A	N/A
10426692	12/2018	Jurna et al.	N/A	N/A
D868277	12/2018	Yamazaki	N/A	N/A
D869673	12/2018	Chen	N/A	N/A
D873418	12/2019	Scott	N/A	N/A
D874666	12/2019	Matsushita	N/A	N/A
D887572	12/2019	Matsushita	N/A	N/A
D890353	12/2019	Nazarian	N/A	N/A
D890941	12/2019	Gaquiere et al.	N/A	N/A
D890943	12/2019	Wersland et al.	N/A	N/A
D894414	12/2019	Sherrill	N/A	N/A
D895823	12/2019	Khubani et al.	N/A	N/A
D896393	12/2019	Wersland et al.	N/A	N/A
10765199	12/2019	Kern	N/A	A61H 7/003
10772473	12/2019	Johnstone	N/A	A47K 7/04
D901032	12/2019	Gonglach	N/A	N/A
D903140	12/2019	Andrejs	N/A	N/A
D905262	12/2019	Khubani et al.	N/A	N/A
D907792	12/2020	Marton et al.	N/A	N/A
D908235	12/2020	Marton et al.	N/A	N/A
D918404	12/2020	Wersland et al.	N/A	N/A
D918405	12/2020	Wersland et al.	N/A	N/A
D925128	12/2020	Yang et al.	N/A	N/A
D927722	12/2020	Park	N/A	N/A
11110029	12/2020	Xie et al.	N/A	N/A
D941036	12/2021	Hinds	N/A	N/A
D945076	12/2021	Zhang	N/A	N/A
D946166	12/2021	Li	N/A	N/A
D949362	12/2021	Zheng	N/A	N/A
D949365	12/2021	Li	N/A	N/A
D949366	12/2021	Li	N/A	N/A
D949374	12/2021	Black	N/A	N/A
D949380	12/2021	Khubani et al.	N/A	N/A
D949395	12/2021	Lee et al.	N/A	N/A
D950062	12/2021	Vuillemin et al.	N/A	N/A
D952166	12/2021	Peluso et al.	N/A	N/A
D952175	12/2021	Zou	N/A	N/A
D953555	12/2021	Lee et al.	N/A	N/A
D954285	12/2021	Lee et al.	N/A	N/A

D956240	12/2021	Mcbrien et al.	N/A	N/A
D956996	12/2021	Costa	N/A	N/A
D961795	12/2021	Zhao et al.	N/A	N/A
D962556	12/2021	Wu	D30/121	N/A
D967978	12/2021	Liu	N/A	N/A
D975045	12/2022	Xu	N/A	N/A
D977667	12/2022	Yang	N/A	N/A
D977842	12/2022	Lee	N/A	N/A
D979087	12/2022	Yang	N/A	N/A
D981578	12/2022	Yee	N/A	N/A
D982356	12/2022	Kliegman et al.	N/A	N/A
D984977	12/2022	Sun	N/A	N/A
D985147	12/2022	Zhu	N/A	N/A
D985786	12/2022	Yin	N/A	N/A
D992131	12/2022	Marton	N/A	N/A
D992132	12/2022	Marton	N/A	N/A
D992135	12/2022	Omari et al.	N/A	N/A
D993437	12/2022	Klimov	N/A	N/A
D994131	12/2022	Black	N/A	N/A
D994132	12/2022	Zong	N/A	N/A
D994140	12/2022	Yang	N/A	N/A
D995808	12/2022	Yin	N/A	N/A
D996065	12/2022	Lei et al.	N/A	N/A
D998149	12/2022	Ghoniem	N/A	N/A
D998160	12/2022	Shao	N/A	N/A
D1003313	12/2022	Lin	N/A	N/A
D1004114	12/2022	Black	N/A	N/A
D1005500	12/2022	Yoon et al.	N/A	N/A
D1005516	12/2022	Leng	N/A	N/A
D1005692	12/2022	Russo	N/A	N/A
D1006247	12/2022	Qiao	N/A	N/A
D1021989	12/2023	Tsushima et al.	N/A	N/A
D1026228	12/2023	Black	N/A	N/A
D1038434	12/2023	Krut	N/A	N/A
D1061915	12/2024	Black	N/A	N/A
2002/0147418	12/2001	Huang	N/A	N/A
2002/0151930	12/2001	Mills	N/A	N/A
2002/0193715	12/2001	Slack	N/A	N/A
2003/0158505	12/2002	Calvert	N/A	N/A
2003/0212353	12/2002	Kahn	N/A	N/A
2004/0024336	12/2003	Lin	N/A	N/A
2004/0082449	12/2003	Brown, Jr.	N/A	N/A
2004/0208683	12/2003	Shawan et al.	N/A	N/A
2004/0260215	12/2003	Kim	N/A	N/A
2005/0015030	12/2004	Bousfield et al.	N/A	N/A
2005/0096572	12/2004	Hua	N/A	N/A
2005/0101944	12/2004	Williams	N/A	N/A
2005/0107724	12/2004	Dunmore	N/A	N/A
2005/0137505	12/2004	Munday	N/A	N/A
2005/0142093	12/2004	Skover	601/84	A61Q 19/08

2006/0135323	12/2005	Sjobakk et al.	N/A	N/A
2006/0253054	12/2005	Wright	N/A	N/A
2007/0173750	12/2006	Hudock	N/A	N/A
2007/0179412	12/2006	Imboden	601/72	H02J 50/12
2008/0086066	12/2007	Munday	N/A	N/A
2008/0119913	12/2007	Powell	606/9	A61N 5/0616
2008/0167590	12/2007	Jon et al.	N/A	N/A
2008/0200849	12/2007	Hollington et al.	N/A	N/A
2008/0255486	12/2007	Ludlow	N/A	N/A
2009/0121059	12/2008	Rios Garcia	N/A	N/A
2009/0204061	12/2008	Pomposelli et al.	N/A	N/A
2009/0287195	12/2008	Altshuler et al.	N/A	N/A
2010/0040432	12/2009	Totsu	N/A	N/A
2010/0160841	12/2009	Wu	N/A	N/A
2010/0191161	12/2009	Mouatt	N/A	N/A
2010/0217164	12/2009	Meyer et al.	N/A	N/A
2010/0274162	12/2009	Evans	N/A	N/A
2010/0312158	12/2009	Lin	N/A	N/A
2011/0054369	12/2010	Destefano	N/A	N/A
2011/0077561	12/2010	Choly	N/A	N/A
2011/0087140	12/2010	Lee	N/A	N/A
2011/0144550	12/2010	Suttman	N/A	N/A
2011/0154690	12/2010	Walsh	N/A	N/A
2011/0224588	12/2010	Grippo	N/A	N/A
2012/0016292	12/2011	Goldberg et al.	N/A	N/A
2012/0109043	12/2011	Zhou et al.	N/A	N/A
2012/0121313	12/2011	Thiebaut	N/A	N/A
2012/0167323	12/2011	Williams	N/A	N/A
2012/0204369	12/2011	Watanabe et al.	N/A	N/A
2012/0323151	12/2011	Faussett	N/A	N/A
2013/0023807	12/2012	Hennessey	N/A	N/A
2013/0046212	12/2012	Nichols	601/18	A61N 1/00
2013/0079689	12/2012	Thierman	601/46	A61H 7/00
2013/0190664	12/2012	Johnson	N/A	N/A
2013/0197405	12/2012	Williams et al.	N/A	N/A
2013/0226052	12/2012	Niggemann	N/A	N/A
2013/0324792	12/2012	Mizrahi et al.	N/A	N/A
2014/0200495	12/2013	Jones	N/A	N/A
2014/0243718	12/2013	Black	N/A	N/A
2014/0272761	12/2013	Lowe et al.	N/A	N/A
2014/0323993	12/2013	Wilcox et al.	N/A	N/A
2014/0358045	12/2013	Toto	N/A	N/A
2015/0135455	12/2014	Reusche et al.	N/A	N/A
2015/0157414	12/2014	Van Curen	N/A	N/A
2015/0216292	12/2014	Tweel et al.	N/A	N/A
2016/0008213	12/2015	Cheng	N/A	N/A
2016/0099595	12/2015	Chien	N/A	N/A
2016/0175185	12/2015	Buchner Santos et al.	N/A	N/A

2016/0206087	12/2015	Skidmore	N/A	A46B 15/0004
2017/0073050	12/2016	Smith	N/A	B08B 1/36
2017/0216136	12/2016	Gordon	N/A	N/A
2017/0333280	12/2016	Black	N/A	N/A
2017/0348082	12/2016	Ogleby	N/A	A01K 13/00
2018/0161233	12/2017	Nakanishi	N/A	N/A
2018/0288160	12/2017	Paul et al.	N/A	N/A
2019/0045912	12/2018	Mikitovic et al.	N/A	N/A
2019/0045918	12/2018	Przirembel et al.	N/A	N/A
2019/0216678	12/2018	Yang	N/A	N/A
2019/0247268	12/2018	Black	N/A	N/A
2020/0113322	12/2019	Balestrini	N/A	N/A
2021/0100717	12/2020	Beck	N/A	N/A
2021/0299304	12/2020	Concannon et al.	N/A	N/A
2022/0087893	12/2021	Black	N/A	N/A
2022/0241136	12/2021	Costa	N/A	N/A
2023/0010191	12/2022	Black	N/A	N/A
2023/0172799	12/2022	Black	N/A	N/A
2023/0200811	12/2022	Black	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
655910	12/1994	AU	N/A
2438449	12/2000	CN	N/A
2449536	12/2000	CN	N/A
102247277	12/2010	CN	N/A
202802101	12/2012	CN	N/A
203694035	12/2013	CN	N/A
203724434	12/2013	CN	N/A
204484727	12/2014	CN	N/A
201530461899.2	12/2015	CN	N/A
201630416937.7	12/2016	CN	N/A
207785468	12/2017	CN	N/A
208626154	12/2018	CN	N/A
208770335	12/2018	CN	N/A
109820708	12/2018	CN	N/A
209464314	12/2018	CN	N/A
209864569	12/2018	CN	N/A
210750164	12/2019	CN	N/A
211326656	12/2019	CN	N/A
108430426	12/2019	CN	N/A
212416317	12/2020	CN	N/A
112426341	12/2020	CN	N/A
213373701	12/2020	CN	N/A
213431541	12/2020	CN	N/A
214049658	12/2020	CN	N/A
307097258	12/2021	CN	N/A
29710313	12/1996	DE	N/A

20 2009 014 825	12/2009	DE	N/A
003539014-0001	12/2016	EM	N/A
0 811 365	12/1996	EP	N/A
2 311 426	12/2010	EP	N/A
2 922 518	12/2014	EP	N/A
2 331 706	12/1998	GB	N/A
2002-153531	12/2001	JP	N/A
6793200	12/2019	JP	N/A
100941568	12/2009	KR	N/A
300888480	12/2016	KR	N/A
2070018	12/1995	RU	N/A
2662879	12/2017	RU	N/A
2719940	12/2019	RU	N/A
201412300	12/2013	TW	N/A
WO-87/00425	12/1986	WO	N/A
WO-02/083054	12/2001	WO	N/A
WO-2007/051233	12/2006	WO	N/A
WO-2008/049165	12/2007	WO	N/A
WO-2012/009862	12/2011	WO	N/A
WO-2014/118596	12/2013	WO	N/A
WO-2017/193054	12/2016	WO	N/A

OTHER PUBLICATIONS

Amazon, “Mini Paddle Blaster”, Date First Available: Oct. 27, 2020. https://www.amazon.com/Mini-Paddle Blaster-Cellulite-FasciaBlaster/dp/B08LZR7GWW/ref=cm_cr_arp_d_project_top?ie=UTF8 (Year: 2020). cited by applicant
Amazon, “Savage Blaster” by Ashley Black, Earliest Review Date: Feb. 24, 2022. https://www.amazoon.com/FasciaBlaster-SavageBlaster-Ashley-Black-Discomfort/dp/B09KM8MWX3/ref=cm_cr_arp_d_pl_foot_top?ie=UTF8 (Year: 2022). cited by applicant
Ashleyblackguru, “Ashley Black Mini Paddleblaster”, Earliest review Date: Aug. 21, 2020. https://www.ashleyblackguru.com/product/mini-paddleblaster (Year: 2020). cited by applicant
Ashleyblackguru, “Savage Blaster”, Earliest Review Date: Nov. 12, 2021. https://www.ashleyblackguru.com/collections/body-tools/products/savage-blaster#shopify-product-reviews (Year: 2021). cited by applicant
International Search Report and Written Opinion on PCT/US2022/052052 dated Mar. 10, 2023, 12 pps. cited by applicant
Youtube, “Discover the SavageBlaster”, Published Date: Jul. 22, 2022I https://www.youtube.com/watch?v=uBxEX3waWSk (Year 2022). cited by applicant
Youtube, “How To Select The Right Fascia Blaster For You”, Published Date: Nov. 27, 2020. https://www.youtube.com/watch?v=bQojqSYrhwQ (Year: 2020). cited by applicant
Amazon, “FasciaBlaster Trigger Point Tools”, Nov. 6, 2017, https://www.amazon.com/Mini-Blaster-Ashley-Black-Guru/dp/B075M26CJT/ref=cm_cr_arp_d_pl_foot_top?ie=UTF8. Shown on p. 1. (Year: 2017). cited by applicant
Amazon, “Mini Paddle Blaster”, Apr. 8, 2021. https://www.amazon.com/Mini-Paddle-Blaster-Cellulite-FasciaBlaster/dp/B08LZR7GWW/ref=sr_1_2?dchild=1&keywords-Ashley+Black+Guru&qid=1617887539&sr=8-2. Shown on p. 1. (Year: 2021). cited by applicant
Amazon, “Trigger Point Massager Massage Tool”, Jan. 18, 2019. https://www.amazon.com/Trigger-Point-Massage-

Tool/dp/B075FXVB39/ref=cm_cr_arp_d_product_top?ie=UTF8. Shown on p. 1. (Year: 2019). cited by applicant
Ashley Blacks, “Mini Paddleblaster1”, Aug. 21, 2020. https://www.ashleyblackguru.com/products/mini-paddleblaster-1#shopify-product-reviews. Shown on p. 1. (Year: 2020). cited by applicant
Ashleyblackguru, “Mini 2”, Nov. 12, 2017, https://www.ashleyblackguru.com/products/mini-2#shopify-product-reviews. Shown on p. 1. (Year: 2017). cited by applicant
Communication pursuant to Rules 161 (1) and 162 EPC for corresponding EP Application No. 14709121.9, mailed Oct. 6, 2015, 2 pages. cited by applicant
D'arsonval High Frequency Device Skin Tightening Spot Wrinkles Remover Acne Treatment, www.topbeautybuy.com, 2017, 6 pages. cited by applicant
European Pat. Appl. No. 14709121.9, Communication dated Feb. 9, 2021, 6 pages. cited by applicant
Examination Report for corresponding EP Application No. 14709121.9, mailed Dec. 7, 2017, 5 pages. cited by applicant
Fascia Blaster Cellulite Massager Full Body Massage Us Seller Lot of 2, www.terapeak.com, 4 pages. cited by applicant
International Preliminary Report on Patentability for corresponding PCT/US2014/018043, mailed Aug. 25, 2015, 6 pages. cited by applicant
International Preliminary Report on Patentability for corresponding PCT/US2017/031384, mailed Nov. 6, 2018, 6 pages. cited by applicant
International Search Report and Written Opinion for corresponding PCT/US2014/018043, mailed May 6, 2014, 10 pages. cited by applicant
International Search Report and Written Opinion for corresponding PCT/US2017/031384, mailed Sep. 14, 2017, 9 pages. cited by applicant
Vola Massager, Wayback Machine Jan. 17, 2008 capture of http://www.chinatraderonline.com/Massager/VOLA-MASSAGER-193802857.htm, retrieved from https://web.archive.org/web/20080117105426/http:/www.chinatraderonline.com/Massager/VOLA-MASSAGER-193802857.htm, Jul. 19, 2021, 1 pg. cited by applicant
Youtube, “Mini1 Paddleblaster”, Oct. 9, 2020. https://www.youtube.com/watch?v=LIK17dCevBQ. Shown at the 0:02 second mark. (Year: 2020). cited by applicant
'057 Patent Holder Disclosure “Product Development LumpBuster Case Study” (LB Case) date May 26, 2013, pdf archive link: https://web.archive.org/web/20111228161639/http://www.chinatraderonline.com/Beauty-Equipment/Octopus-Massager-13342511.htm, 2 pgs. cited by applicant
Amazon, “FasciaBlaster ProBlaster”, Earliest Review Date: Oct. 14, 2022. https:/www.amazon.com/FasciaBlaster-ProBlaster-Ashley-Black-Discomfort/dp/BOB916ZPYP/ref=cm_cr_arp_d_pl_foot_top?ie=UTF8 (Year: 2022). cited by applicant
Ashleyblackguru, “Problaster”, Earliest Review Date: Aug. 9, 2022. https:/Avww.ashleyblackguru.com/collections/all/products/problaster (Year: 2022). cited by applicant
Definition of the term “Circular”, Miriamwebster.com, captured on Feb. 1, 2024. cited by applicant
Octopus-Shaped Massager (Octopus), Sales Pgs Dated 2006-2011, Retrieved from: https://web.archive.org/web/20060514183134/http://www.chinatraderonline.com/Files/Health-2Beauty/Beauty-Equipment/Octopus-Massager-13342022084.jpg, 1 pg. cited by applicant
“BioSwiss 2-Pack Full Body Massager”, Sep. 29, 2020, Site visited Apr. 15, 2025, Available from Internet URL:https://www.amazon.com/dp/B07X8K9N61/?th=1. cited by applicant
“The Single Claw FasciaBlaster by Ashley Black”, Dec. 16, 2022, Site visited Apr. 15, 2025,

Available from Internet URL:https://www.amazon.com/Singel-Claw-FasciaBlaster-Ashley-Black/dp/BOBM4VPVYV?ref_=ast_sto_dp. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) The present application is a Continuation-in-Part of co-pending U.S. patent application Ser. No. 17/545,920, filed on Dec. 8, 2021, which claims priority to, and the benefit of, U.S. Provisional Patent Application Ser. No. 63/286,536, filed on Dec. 6, 2021, and further claims priority to, and the benefit of, U.S. Provisional Patent Application Ser. No. 63/347,961, filed Jun. 1, 2022, the contents of which are hereby incorporated herein by reference in their entirety.

BACKGROUND

(1) Fascia tissue is a layer of fibrous tissue that operates as a connective tissue that surrounds muscles, groups of muscles, nerves, blood vessels, etc. The tissue allows for proper functioning of muscles with respect to one another (e.g., sliding past one another). When fascia tissue becomes damaged through injury, tissue knots, adhesions in the fascia tissue, medical reasons, or otherwise, the fascia tissue can take time to correct itself or require manipulation to release the fascia tissue and allow for proper functioning of the tissue to allow the underlying muscle to properly operate. In some cases, the fascia tissue can be released or corrected without much difficulty, while in other cases, restoring the fascia tissue to its proper form can take considerably more effort. Other reasons for releasing fascia tissue may include cosmetic reasons, especially for people who have dimpled skin, which is often caused by fascia tissue extending through fat cells, thus causing dimples to appear on the skin. Often, when the fascia tissue is properly released, the dimples can be considerably reduced or eliminated.

(2) Fascia tissue is both structural and inter-structural. Fascial tissue is located throughout the human (and other animals) bodies, including attaching to skin, forming a casing over muscles, forming inter-muscular layers (myofascial tissue), being positioned at lower depths of layered muscles, surrounding organs, and located around bones. For the lower depth fascia, especially when multiple muscles stacked are stacked, adhesions of the fascia tissue or other fascia tissue structure misalignments or damage can result in a variety of discomforts or worse problems, such as circulatory flow, pain, misalignment of muscles and bones, joint problems, and so forth. As such, there is a need to treat fascia tissue at any location of the body.

(3) Fascia tissue treatment is complicated and scientifically challenging to diagnose and treat. The diagnosis and understanding of a myriad of problems that result from damaged or misaligned fascia tissue is not well understood, and, thus, often not properly treated. For example, pain, circulatory problems, and other physical ailments are often treated using pain medicine, surgery, heating pads, conventional massage, chiropractor join adjustment, and many other medical treatments that essentially treat the symptom and not the cause of the physical ailments that stem from damaged or misaligned fascia tissue. Moreover, even if diagnosed, the ability to treat the fascia tissue, especially myofascial tissue, can be difficult because tools to treat such tissue are limited and generally not configured for such treatments.

SUMMARY

(4) To overcome the problem of having limited tools for treating fascia tissue, the principles described herein provide for treating a wide variety of fascia tissue that may be positioned

throughout the body, human and animal. The tools described herein are configured to treat or remodel (e.g., removal of adhesions, realignment, etc.) fascia tissue at depths and orientations that cannot be achieved using other tools to massage techniques. For example, rollers, nubs, or devices with other configurations generally are not capable of penetrating muscle casing or surface of a muscle much less layered muscle (e.g., two or three layers of muscle of quadriceps muscles). To property treat the fascia tissue, powered (and unpowered) devices with effectors (e.g., fingers of a certain configuration connected to a panel) creates a “fascial shearing” that triggers the body's natural mechanism to remodel or restructure the fascia tissue that has been damaged or is disfigured in a manner that results in negative effects on the human body.

(5) A fascia tissue treatment system may include a base station that powers tissue treatment devices that are used to treat fascia or other tissue of patients. The tissue treatment devices may be powered devices, and include actuators that are used to move tissue treatment elements. The actuators may be motors that cause an effector inclusive of a support structure, such as a disc or other shaped structural member, on which the tissue treatment elements are connected or formed. In an embodiment, the effector may be a unitary piece, where a support structure and protrusions or fingers extend. Alternatively, the fingers may be detachably coupled to the support structure. The fingers may have a base that connects to or is integrated with the support structure, curved shafts, and rounded tips.

(6) In at least one embodiment, the tissue treatment elements may form part of a tissue treatment assembly that is configured for rotation by the actuator in an ON state. The tissue treatment assembly may be uniquely designed to improve treatment effectiveness based on the direction of movement of the actuator. For example, the tissue treatment assembly may include a panel and a plurality of finger members that are shaped based on a rotational direction of the actuator or panel. In at least one implementation, the finger members are curved away from the panel along at least one of a radial direction and/or a circumferential direction of the panel, which can also increase the strength of the finger members during operation and in response to application of axial pressure acting on the finger members during treatment.

(7) In operation, base station may be utilized to provide electrical power (or other power source types) to provide tissue treatment functions, such as warm-up, tissue release, and/or tissue treatment. The base station may support (i) a heater, such as an infrared heater element, to warm up tissue of a patient, (ii) an ultrasound and/or radio frequency (RF) device to provide for tissue release, and tissue treatment devices of different configurations to provide for fascia or other tissue treatment.

(8) Moreover, the tissue treatment device may receive electrical power that drives a rotating motor therein. The motor may cause the effector to spin. An operator, either a medical professional or a patient him or herself, may engage the tissue treatment device to cause the effector to spin while pressing the effector onto a patient's skin. The amount of force being applied may vary depending on a modality being performed along with a particular location on the patient's body being treated. For example, treatment to a face or scalp uses much less force than body parts with larger surface area and more dense tissue (e.g., muscle). Moreover, if fascia tissue being treated is causing acute pain, then less force and, possibly, less speed of the rotating effector may be used. It should be understood that a wide variety of treatment protocols may be utilized. Moreover, although a spinning effector may be utilized, other actuator types and effectors may be utilized. For example, motors that cause an effector, such as a square or rectangular effector, to cause the effector to move linearly (e.g., forward and backward along a single axis). Powered tissue treatment devices may be utilized, as well, and have a variety of different controllers.

(9) The base station may further include an electronic device, such as a tablet, that may be used to manage operators, patients, treatment plans, collect data, and/or control operations of the tissue treatment device(s). In an embodiment, the base station may further be configured with an ultrasound scanning device and optionally camera device that enables an operator to perform

ultrasound scanning on tissue of a patient and record the scans to aid the operator know the type of treatment that the patient in that region should have. The computing device may capture the ultrasound images and/or visual images and store those images for later viewing. Additionally, the computing device may compare the images over time to assist an operator in determining how past treatments have resulted in improving tissue of the patient. In an embodiment, artificial intelligence (AI) software may be utilized to identify fascia tissue or other tissue abnormalities or structural issues to aid an operator understand how treatments are to be made. In an embodiment, the computing device may be configured to recognize certain structural issues with fascia tissue and aid or recommend to an operator or other medical professional in creating a treatment plan.
(10) One embodiment of the present disclosure relates to a powered treatment device for treating fascia tissue. The powered treatment device may include a housing, an actuator, and a tissue treatment assembly. The actuator may be configured to rotate and may be coupled to and disposed substantially within the housing. The tissue treatment assembly may include a panel and a plurality of finger members. The panel may be detachably coupled to the actuator so as to be rotated with respect to the housing when the actuator is in an ON state. The plurality of finger members may be fixedly coupled to the panel at a proximal end of the respective finger members. The finger members may be rigid and extend away from the panel. Each one of the finger members may have a respective central axis that extends from the proximal end to a distal end of the respective finger member. At least one of the central axes may extend at least partly along a radial direction and/or a circumferential direction of the panel.
(11) Another embodiment of the present disclosure relates to a tissue treatment assembly that includes a panel, a support member, and a plurality of finger members. The support member may be disposed on the panel and may include a connecting element configured to detachably couple the support member to an actuator. The plurality of finger members may be coupled to the panel at a proximal end of the finger members. The finger members may be rigid and may extend away from the panel. Each one of the plurality of finger members may have a respective central axis extending from the proximal end to a distal end of the respective finger member. At least one of the central axes may extend at least partly along a radial direction and/or a circumferential direction of the panel.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) A more complete understanding of the method and apparatus of the present invention may be obtained by reference to the following Detailed Description when taken in conjunction with the accompanying Drawings wherein:
(2) FIG. **1** is an illustration of a front isometric view of an illustrative fascia tissue therapy system;
(3) FIG. **2** is an illustration of a fascia tissue fitness device of the illustrative therapy system of FIG. **1**;
(4) FIG. **3**A is a rear view of the illustrative fascia tissue therapy system of FIG. **1**;
(5) FIG. **3**B is a side view of the illustrative fascia tissue therapy system of FIG. **1**;
(6) FIG. **3**C is a front view of the illustrative fascia tissue therapy system of FIG. **1**;
(7) FIG. **4** is a rear view of the illustrative fascia tissue therapy system of FIG. **1** shown with an access panel in a closed position;
(8) FIG. **5** is a rear view of the illustrative fascia tissue therapy system of FIG. **1** shown with an access panel in an opened position;
(9) FIG. **6** is a side partial view of the illustrative fascia tissue therapy system of FIG. **1** shown with an access panel in a closed position;
(10) FIG. **7** is a side partial view of the illustrative fascia tissue therapy system of FIG. **1** shown

with an access panel in a partially open position;
(11) FIG. **8** is a side partial view of the illustrative fascia tissue therapy system of FIG. **1** shown with an access panel in a fully open position;
(12) FIG. **9** is an illustration of a front isometric view of another illustrative fascia tissue therapy system;
(13) FIG. **10** is an illustration of a front isometric view of another illustrative carousel for a fascia tissue therapy system;
(14) FIG. **11** is an illustration of a front isometric view of an illustrative carousel for a fascia tissue therapy system;
(15) FIG. **12** is an illustration of a front isometric view of another illustrative carousel for a fascia tissue therapy system;
(16) FIG. **13** is an illustration of a rear view of another illustrative fascia tissue therapy system;
(17) FIG. **14**A is an illustration of a side isometric view of a display mount portion of an illustrative fascia tissue therapy system;
(18) FIG. **14**B is another illustration of a side isometric view of the display mount portion of FIG. **14**A;
(19) FIG. **15**A is an illustration of a front isometric view of another illustrative fascia tissue therapy system;
(20) FIG. **15**B is a rear view of the illustrative therapy system of FIG. **15**A;
(21) FIG. **16**A is an illustration of a front isometric view of yet another illustrative fascia tissue therapy system;
(22) FIG. **16**B is a rear view of the illustrative therapy system of FIG. **16**A;
(23) FIG. **17** is an illustration of a front isometric view of yet another illustrative fascia tissue therapy system;
(24) FIG. **18** is an illustration of a front isometric view of yet another illustrative fascia tissue therapy system;
(25) FIG. **19** is an illustration of a front isometric view of yet another illustrative fascia tissue therapy system;
(26) FIG. **20** is an illustration of a front isometric view of yet another illustrative fascia tissue therapy system;
(27) FIG. **21** is an illustration of a side view of illustrative fascia tissue fitness device of FIG. **2**;
(28) FIG. **22** is an illustration of a side view of another illustrative fascia tissue fitness device;
(29) FIGS. **23**A-**23**D are illustrations of different sides of yet another illustrative fascia tissue fitness device;
(30) FIGS. **24**A-**24**B are side isometric views of yet another illustrative fascia tissue fitness device;
(31) FIG. **25**A is an illustration of an exploded view of an illustrative fascia tissue treatment device;
(32) FIG. **25**B is an illustration of an exploded view of the illustrative fascia tissue treatment device of FIGS. **23**A-**23**D;
(33) FIG. **26**A is an illustration of a front isometric view of another illustrative fascia tissue treatment device;
(34) FIG. **26**B is another front isometric view of the illustrative fascia tissue treatment device of FIG. **26**A;
(35) FIG. **27** is an illustration of a storyboard of a video demonstration for using a therapy device;
(36) FIG. **28** is an illustration of a front isometric view of yet another fascia tissue treatment device;
(37) FIG. **29** is an illustration of a front isometric view of a handle for a fascia tissue treatment device;
(38) FIG. **30** is a front view of the fascia tissue treatment device of FIG. **28**;
(39) FIG. **31** is a side view of the fascia tissue treatment device of FIG. **28**;
(40) FIGS. **32-35** are illustrations of various illustrative handle designs for a fascia tissue treatment device;

(41) FIG. **36** is an illustration of a front isometric view of yet another illustrative fascia tissue treatment device;
(42) FIG. **37** is another isometric view of the illustrative fascia tissue treatment device of FIG. **36**;
(43) FIG. **38** is an illustration of a side isometric view of yet another illustrative fascia tissue treatment device;
(44) FIG. **39** is an illustration of a side isometric view of yet another illustrative fascia tissue treatment device;
(45) FIG. **40**A is an illustration of a cross-sectional view of illustrative fascia tissue layers;
(46) FIG. **40**B is an illustration of a cross-sectional view of a leg showing illustrative structural and inter-structural fascia tissue and muscle;
(47) FIGS. **41-44** are images from an ultrasound analysis of patient tissue taken at different intervals during treatment using a fascia tissue therapy device;
(48) FIG. **45** is a flow diagram of an illustrative method for fascia tissue analysis;
(49) FIG. **46** is an illustration of a bottom view of yet another illustrative fascia tissue treatment device;
(50) FIG. **47** is an illustration of a bottom view of yet another illustrative fascia tissue treatment device;
(51) FIGS. **48-49** are illustrations of bottom views of yet another illustrative fascia tissue treatment device;
(52) FIG. **50** is an illustration of a side view of the illustrative fascia tissue treatment device of FIG. **48**;
(53) FIGS. **51-53** are illustrations of bottom views of yet another illustrative fascia tissue treatment device;
(54) FIG. **54** is an illustration of a side view of the illustrative fascia tissue treatment device of FIG. **51**;
(55) FIG. **55** is an illustration of a isometric view of yet another illustrative fascia tissue treatment device;
(56) FIG. **56** is an illustration of a front exploded view of an illustrative preparation device;
(57) FIG. **57** is an illustration of a rear exploded view of an illustrative preparation device;
(58) FIG. **58** is an illustration of an isometric view of a helmet for an illustrative preparation device;
(59) FIG. **59-62** are illustrations of various illustrative fascia tissue treatment devices for use with a therapy system;
(60) FIGS. **63-64** are illustrations of various illustrative fascia tissue treatment devices that may be used to treat smaller body areas;
(61) FIGS. **65**A-**65**D are illustrations of yet another illustrative fascia tissue treatment device that may be used to treat smaller body areas;
(62) FIGS. **66-73** are illustrations of various illustrative fascia tissue treatment devices that may be used to treat smaller body areas;
(63) FIG. **74** is an illustration of a front isometric view of yet another illustrative fascia tissue treatment device;
(64) FIG. **75** is a rear isometric view of the illustrative fascia tissue treatment device of FIG. **74**;
(65) FIG. **76** is a top isometric view of an illustrative tissue treatment effector;
(66) FIG. **77** is an illustration of a bottom isometric view of the illustrative tissue treatment effector of FIG. **76**;
(67) FIG. **78** is a bottom view of the illustrative tissue treatment effector of FIG. **77**;
(68) FIG. **79** is an illustration of another illustrative tissue treatment effector;
(69) FIG. **80** is an illustration of a bottom view of yet another illustrative tissue treatment effector for a fascia tissue treatment device;
(70) FIG. **81** is an illustration of a bottom view of yet another illustrative tissue treatment effector

for a fascia tissue treatment device;
(71) FIG. **82** is an illustration of a side view of yet another illustrative tissue treatment effector;
(72) FIG. **83** is an illustration of a side view of yet another illustrative tissue treatment effector;
(73) FIG. **84** is an illustration of a side view of yet another illustrative tissue treatment effector;
(74) FIG. **85** is an illustration of a side cross-sectional view of yet another illustrative tissue treatment effector;
(75) FIG. **86** is an illustration of a side cross-sectional view of yet another illustrative tissue treatment effector;
(76) FIG. **87** is an illustration of a side cross-sectional view of yet another illustrative tissue treatment effector;
(77) FIGS. **88-91** are illustrations of various illustrative tissue treatment effectors for a fascia tissue treatment device;
(78) FIG. **92**A is an illustration of a side cross-sectional view of yet another illustrative tissue treatment effector;
(79) FIG. **92**B is an illustration of a side cross-sectional view of yet another illustrative tissue treatment effector;
(80) FIG. **92**C is an illustration of an isometric view of a tissue treatment element for a fascia tissue treatment device;
(81) FIGS. **93-100** and **119** are various before and after photographs of patients after using a fascia tissue therapy system;
(82) FIGS. **101**A-**101**F are illustrations of perspective, another perspective, top, side, front, and bottom views of yet another illustrative fascia tissue therapy system;
(83) FIGS. **102**A-**102**F are illustrations of perspective, another perspective, top, side, front, and bottom views of yet another illustrative fascia tissue therapy system;
(84) FIGS. **103**A-**103**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue fitness device;
(85) FIGS. **104**A-**104**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative tissue treatment effector;
(86) FIGS. **105**A-**105**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue fitness device;
(87) FIGS. **106**A-**106**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustration fascia tissue fitness device;
(88) FIGS. **107**A-**107**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue fitness device;
(89) FIGS. **108**A-**108**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue fitness device;
(90) FIGS. **109**A-**109**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue fitness device;
(91) FIG. **110** is an illustration of an isometric view of yet another illustrative fascia tissue treatment device;
(92) FIG. **11** is an illustration of a side cross-sectional view of the illustrative fascia tissue treatment device of FIG. **111**;
(93) FIG. **112** is an illustration of an isometric view of yet another illustrative fascia tissue treatment device;
(94) FIGS. **113-116** are illustrations of side views of yet other illustrative fascia tissue treatment devices;
(95) FIG. **117** is an illustration of a top isometric view of yet another illustrative fascia tissue treatment device;
(96) FIG. **118** is an illustration of a bottom view of an effector of the illustrative fascia tissue treatment device of FIG. **117**;

(97) FIG. **119** are images of a foot having poor circulation before fascia tissue treatment and after fascia tissue treatment; and

(98) FIG. **120** is an illustration of an inter-vaginal fascia tissue treatment device including various treatment effectors that may be driven thereby for performing fascia tissue treatment within a vagina. In an embodiment, the device **12000** may have an adapter configured to receive an extension handle to enable the user to treat herself while laying down.

DETAILED DESCRIPTION

(99) Existing fascia tissue treatment devices require manual manipulation by a user to move tissue treatment elements across a patient's skin. While effective, users performing self-treatment using hand-held tools may not use sufficient force and/or speed to ensure adequate treatment of the fascia tissue. Conventional motorized scalp or tissue massagers are not designed to release fascia tissue. For example, the messaging and/or brush elements employed by these conventional systems are specifically designed for user comfort and skin surface treatment, but are not suitable for working the underlying tissue in a manner meant to improve fascia tissue fitness to perform restructuring of fascia tissue at different layers starting at the skin and working down towards or even to the bone. Additionally, fascia tissue treatment may utilize considerable application of force between the tissue treatment elements and a person or patient's skin, which may cause stress to an actuator being used to drive the tissue treatment elements, thereby resulting in premature failure of the actuator. Such stress, for example, may be caused by using forces in treating fascia tissue that higher torque has to be produced by the actuator, which may result in heat produced in the actuator due to high current draw therein. As such, various configurations of actuators and tissue treatment elements may be utilized to avoid such high-torque situations. It should be understood that the fascia tissue treatments are meant to be applied to lubricated skin of a patient, where the lubrication may be any fluid, such as oil, applied to the skin that reduces friction between tissue treatment elements, such as protrusions (e.g., fingers), of an effector of the tissue treatment device. Alternatively, a cover, such as a sleeve formed of a slick external material, may be placed on skin of a user to enable the tissue treatment device to be applied to the cover to avoid or use a minimal amount of lubrication.

(100) Referring to the Figures generally, a fascia tissue therapy system is shown that addresses the foregoing issues and provides an all-in-one system for diagnosing and treating damage to fascia tissue. The therapy system includes a base station and a therapy device inclusive of an actuator and tissue treatment element that is actuated (e.g., rotated, vibrated, linearly translated, etc.) by the actuator. The tissue treatment element may form part of an effector that is detachably coupled to the base station. The base station may be configured in a number of ways, including (i) operating as a cabinet with actuated heads that control operation of an effector with tissue treatment elements, (ii) operating a computing device to help monitor and manage treatment sessions of patients by an operator of the therapy system, and (iii) operating a computing device that includes a control system for the therapy system that is configured to (a) determine a tissue condition based on data received from the therapy device (e.g., sensors onboard the therapy device, etc.), (b) determine a treatment regimen based on the condition, and (c) control and monitor operation of the therapy device to treat the condition.

(101) Because the fascia tissue therapy system may support different types of treatment devices, such as radio frequency (RF) treatment device(s), ultrasonic cavitation treatment device(s), and fascia tissue treatment device(s), different power types and levels may be utilized for each of the treatment devices. As such, the system may be configured to supply different power types and/or levels to support each of the different devices, and be configured to manually, semi-automatically, or automatically set a different power type and/or level or include different power sources for such support. The different power types and/or levels may be controlled by mechanical, electrical, and/or software and/or hardware (e.g., mechanical switch or other selector) by an operator, for example. In an embodiment, electronics may be utilized to condition power for the different types

and/or levels. For example, if three different power types (e.g., DC 100 v, AC 120 v/1-phase, AC 208 v/3-phase) as to be used to support three different treatment devices, the electronics that supply power each of those type types to one or more outlets at the base station into which an operator may plug the treatment devices.

(102) In an embodiment, the computing system may also be configured to track treatment progress and re-evaluate the tissue condition at periodic (e.g., during periodic treatments) or aperiodic (e.g., during non-periodic treatments) intervals. The therapy device may be designed to monitor and treat damage to fascia tissue or otherwise manipulate and restore fascia tissue. The therapy device may include a direct drive motor and an effector that is detachably coupled to the therapy device and powered into rotation (or otherwise) by the direct drive motor. Unlike existing powered devices, which typically use a transmission (e.g., gear, gear set, etc.) to achieve the high speeds required for operation, the direct drive motor of the present disclosure may be coupled to the effector without an intervening transmission or gear set. Using a direct drive motor may reduce heat generation and wear on the treatment device as compared to a typical motor that utilizes a gear box, thereby allowing use of the device over longer treatment intervals and with less risk of changes in performance during treatment. A high torque-to-weight ratio of the direct drive motor may also provide a lower weight of the therapy device as compared to a therapy device that uses a non-direct drive motor that includes gears, cooling fluid and channels through which the cooling fluid flows, etc.

(103) The effector for the treatment device may be structured to engage a patient's skin and manipulate the fascia tissue to restore and restructure the fascia tissue to its proper form. In being structured to engage a patient's skin, the effector may be shaped to avoid cutting, scraping, or otherwise harm a patient's skin while being configured to treat the fascia tissue beneath the skin. In contrast with conventional motorized scalp or tissue massager designs, the effector and/or tissue treatment assembly of the present disclosure may be shaped or otherwise structured based on a direction of movement of the effector during treatment to improve treatment effectiveness as compared to effector designs that are not oriented with the direction of travel of the effector. As described herein, the system may be used to treat the fascia tissue at different levels (e.g., shallow, medium depth, deep) depending on the modality and location of the patient (e.g., face may be shallow, arm may range from shallow to medium, leg may range from shallow to deep). In some embodiments, the effector may be configured to treat shallow, medium depth, or deep tissue. For example, the shape of the effector and fingers (or other features) that are to contact a patient's skin may have different lengths, widths, tip diameters, curve angles, or other dimensions to be used for shallow, medium depth, or deep tissue. Shallow may mean up to one-tenth of an inch, medium depth may mean up to one inch, and deep tissue may mean more than one inch. It should be understood that alternative ranges may be utilized.

(104) The effector may include a panel (e.g., a support structure, etc.) and/or plate that is detachably coupled to the treatment device and that includes multiple tissue treatment elements. In some embodiments, the panel may include a mount at an intermediate radial position along the panel that connects the panel to the treatment device to reduce a torque required to maintain movement of the panel during operation. The panel may also include supports and/or be shaped to ensure uniform contact between the tissue treatment elements and the patient's skin across the face of the panel. In an embodiment, the panel may be flat, round, and have the protrusions e.g., tissue treatment elements) on one side of the panel. However, it should be appreciated that “panel” as used herein also includes various other support structure shapes. For example, in some embodiments, the panel may be curved or rounded (e.g., parabolic) from a base portion to a tip with the tissue treatment elements connected to and radially extending therefrom. In yet other embodiments, the panel can include other non-flat structures such as a waffle structure, a honeycomb-shaped structure, a dome and/or conical extension and/or protrusion. In further embodiments, the panel may include or be formed from a non-contiguous structure such as ribs,

bars, standoff-bridges, etc.
(105) The tissue treatment elements may include a plurality of rigid (e.g., inflexible, stiff, or not easily bent, etc.) protrusions (e.g., fingers and/or finger members, wave shape, pyramid shape with curved tips, spline, or other geometric or non-geometric shape) that extend axially away from the panel. In further contrast to conventional motorized scalp and tissue massagers, the effectors of the present application may be specifically structured to manipulate fascia tissue layers below the skin surface and to take advantage of the direction of movement of the effector during treatment to improve overall treatment effectiveness. For example, tissue treatment elements (e.g., fingers) may extend away from panel (e.g., flat circular member) at an angle (e.g., along a radial direction and/or a direction of rotation of the effector) to improve treatment efficacy and/or allow different treatment modalities depending on the rotational direction of the effector. The tissue treatment elements (e.g., fingers, member that includes one or more fingers that may be connected to a panel, or any other structure that includes elements that contact skin for treating fascia tissue) may have a variety of different shapes, sizes, orientations, alignments, and/or configurations. In one embodiment, the tissue treatment elements (e.g., fingers) may have multiple different sized fingers (e.g., small in the middle and large on the outside of the panel having a flat, circular shape or any other shape that may or may not be flat).
(106) Referring to FIG. **1** and FIGS. **3**A-**3**C, an illustrative fascia tissue therapy system **100** (e.g., treatment system, etc.) is shown. As shown in FIG. **3**B, the system **100** includes a base station **102** (e.g., cabinet, power source(s), computer, communications equipment, etc.) that is configured to house and/or support various equipment, such RF treatment device(s), ultrasound device(s), a computing system **103**, one or more powered fascia tissue treatment devices **104**, and/or other system components. The base station **102** includes an enclosure **106** defining an interior cavity **108** (as shown in FIG. **5**) that is sized to store multiple fascia tissue treatment devices, tissue treatment assemblies, and/or tissue treatment elements therein. The base station **102** may also include a set of wheels **110** (e.g., casters, rollers, etc.) mounted to a bottom or lower wall **112** of the enclosure **106** to facilitate movement of the base station **102**. The wheels **110** may be rotatably coupled to the base station **102** and may include locks to allow a user to prevent rotation of the wheels **110** once the base station **102** has been moved to a desired location. The number and/or arrangement of wheels **110** may be different in other embodiments.
(107) As shown in FIG. **3**B, the base station **102** includes at least one mount **114** configured to support at least one treatment device **104** in position on the base station **102**. The mount **114** may include a rack that detachably couples or otherwise supports the fascia treatment device **104** on the enclosure. In the embodiment of FIG. **3**B, the mount **114** may include multiple slots **116** (e.g., recessed areas, depressions, etc.) disposed along an outer perimeter of an upper wall **118** or sidewalls **120** of the enclosure **106**. The slots **116** may be sized and configured to receive a handle **122** and/or head portion of the treatment device **104** therein to support the treatment device **104** along the upper end of the base station **102**. In the embodiment of FIG. **1**, the handle **122** hangs on an outer wall of the slot **116** and is secured in position by gravity acting on the treatment device **104**. In other embodiments, the mount **114** may include other types of hangers and/or cradles. In yet other embodiments, the mount **114** may include magnets, hook and loop fasteners, and/or another form of detachable coupling to secure the treatment device **104** to the enclosure **106** when not in use.
(108) FIG. **2** shows an illustrative fascia tissue treatment device **200** (e.g., head, etc.) for the system of FIG. **1**, which may be the same as or similar to the treatment device **104** of FIG. **1**. As shown in FIG. **2**, the treatment device **104**, **200** is electrically coupled to the base station **102** (e.g., a power source that is disposed within the base station **102**) via a tether **124** (e.g., electric cable, etc.) that is coupled to and extends away from the enclosure **106**. The tether **124** also includes electrical connections to power the treatment device **104**, **200** and to transmit signals between sensors of the treatment device **200** and the computing system **103**. In at least one embodiment, the treatment

device **104**, **200** is detachably coupled to the tether **124** so that the same tether **124** can be used to power multiple different treatment devices depending on the desired modality. In an embodiment, all fascia tissue treatment devices **104**, **200** may be configured to utilize the same power signals, thereby enabling the system **100** to include a single power cord to which different tissue treatment devices may connect. Alternatively, different tissue treatment devices may be configured to utilized different electrical signals such that different adapters, power cords, sockets, etc. may be used for the different fascia tissue treatment devices. In yet another embodiment, one or more fascia tissue treatment devices **104**, **200** may be configured with a rechargeable battery or other power source to be used in a cordless manner and be charged by the base station **102**. For example, the base station **102** may be configured to provide power to larger treatment devices **104**, **200**, which may be configured to treat certain anatomical regions (e.g., back, core, legs, arms) to perform medium or deep tissue modalities, for example, may be wired, while smaller treatment devices that are used for other anatomical regions (e.g., face, neck, scalp, hands, feet, etc.) or shallow modalities (e.g., skin, scalp, etc.) may be self-powered.

(109) FIGS. **3**A-**3**C show rear, side, and front views, respectively, of the illustrative base station **102** of FIG. **1** are shown. The enclosure **106** of the base station **102** includes an access panel **126** (e.g., door, etc.) coupled to the enclosure **106** along a rear side **128** of the enclosure **106**. The access panel **126** is movable between an open position and a closed position to selectively provide access to the interior cavity **108** of the enclosure **106** (as shown in FIG. **5**). FIGS. **4-8** show various views of the illustrative base station **102** of FIG. **1** with the access panel **126** in the open position **130** (FIG. **8**) and the closed position **132** (FIG. **6**). As shown in FIGS. **6-8**, the access panel **126** includes a pivot system **134** that allows the access panel **126** to translate away from (e.g., radially away from) and pivot with respect to the enclosure **106**. It should be understood that a variety of configurations for opening and closing the access panel **126** may be utilized, including sliding open and close within a concealed wall. The pivot system **134** may include a shock absorber **136** to stabilize the supports **138** for the access panel **126** and facilitate movement between the open position, the closed position, and/or an intermediate position between the open and closed positions. In some embodiments, the pivot system **134** may be configured to automatically reposition the access panel **126** in response to user commands from the computing system **103**.

(110) In some embodiments, the system **100** (e.g., the base station **102**) is structured to store treatment devices, tissue treatment assemblies, and/or tissue treatment elements therein. For example, FIG. **9** shows an illustration of a rear perspective view of another illustrative fascia tissue therapy system **900** that includes a rotating carousel **905**. As shown in FIG. **9**, the rotating carousel **905** is disposed within the interior cavity **904** of the enclosure **906** and is rotatably coupled to the enclosure **906** (e.g., is rotatable within the enclosure **906**). The access panel **926** is moveably coupled to the enclosure **906** and can rotate away from the enclosure **906** to provide access to the carousel **905**. As shown in FIG. **10**, the carousel **905** may include a plurality of mounts **914** arranged circumferentially (e.g., along a circumferential direction) about a rotational axis **916** of the carousel **905**. The mounts **914** are engageable with the treatment device, tissue treatment assemblies, tissue treatment element, and/or other add-ons for the therapy system **900**. The mounts **914** may include hangers, magnets, and/or other detachable couplings to support any add-on equipment for the system **900**. For example, FIG. **11** shows an illustration of a carousel **1100** that includes wire hangers **1102** and shelving **1104** to support various add-ons (e.g., therapy devices, etc.). FIG. **12** shows a carousel **1205** that includes a plurality of loops **1202** extending radially away from a central sleeve **1204**. It will be appreciated that the design of the carousel **1205** may be different in other embodiments. For example, FIG. **13** shows an enclosure **1306** of a base station **1302** that includes a straight rack of hangers **1314** that may be rotatably or fixedly coupled to the enclosure **1306**. As shown in FIG. **15**, the rack of hangers **1314** is arranged along a single row (e.g., a linear row, etc.) within the interior cavity **1308** of the enclosure **1306**. Outlet(s) to power source(s) may be disposed within or outside of the enclosure **1306** and be easily accessible to plugs

of cables for the treatment devices to be plugged into to draw power for operation thereby. In an embodiment, the outlet(s) may be rotatable on the carousel **1305**.

(111) It will be appreciated that different mounts may be used to store different types of equipment within or external to the enclosure. The mounts may be arranged at different positions within the interior cavity and/or along the carousel depending on the type of equipment that the mounts support (e.g., at different axial and/or radial positions along the carousel, etc.). A user may rotate the carousel within the interior cavity and select the piece of equipment that the user would like to use. Beneficially, using a rotatable carousel to store treatment devices and add-ons improves space utilization of the interior cavity.

(112) In the embodiment of FIGS. **9** and **10**, the carousel **905** defines an upper wall **918** of the enclosure **906** that is accessible from an upper end **940** of the enclosure **906**. The carousel **905** may include a plurality of electrical connectors **942** along the upper wall **918** that are engageable with the tether **924** for the treatment device(s). As such, rotation of the carousel **905** may also rotate external features of the base station **902** and/or enclosure **906** (e.g., the upper wall **918**). In other embodiments, as shown in at least FIGS. **1-2** and FIGS. **3**A-**3**C, the position of the electrical connection **146** between the tether **124** and the base station **102** may be different (e.g., the tether **124** may connect to the base station **102** at a lower, rear end of the enclosure **106**, etc.).

(113) As shown in FIG. **13**, a user may store the treatment device **1304** in the interior cavity **1308** when not in use to improve the overall aesthetic of the system **1300**. In some embodiments, the base station **1302** may further include a motor **1340** coupled to the carousel **1305** and structured to rotate the carousel **1305** in response to user commands (e.g., signals from a control interface of the base station, push-button, etc.). In an embodiment, rather than using a carousel **1305**, the mounts may be positioned on the walls of the interior of the enclosure **1306**.

(114) In some embodiments, the base station also includes a sterilization system **1342** configured to sterilize treatment devices **1304**, tissue treatment elements, and/or other equipment. The sterilization system **1342** may include an ultraviolet (UV) light source including one or more UV lights disposed within the interior cavity **1308** and configured to expose or “flood” equipment mounted to the carousel **1302** with UV light for a threshold period after the access panel has been closed (e.g., when the access panel is in a closed position blocking access to the interior cavity **1308**). In an embodiment, the UV light may be controlled to be turned ON or OFF by a push-button, controlled to be turned ON or OFF via a user interface of a computing device, automatically turned ON when the system **1300** is not in use (e.g., overnight), or automatically turned ON when the access panel is closed (monitored by a switch, for example). Alternatively, or in combination, the sterilization system **1342** may include a dispensing system **1344** (e.g., nozzle, pump, etc.) disposed within the interior cavity **1308** and configured to distribute a sterilization agent over the equipment placed within a container (e.g., mounted to the floor or wall) of the sterilization system **1342**, where a portion of a treatment device **1304** may be placed into the container within the interior cavity **1308** in response to a signal indicating that the access panel has been closed, and/or a signal indicating that treatment has concluded. In some embodiments, the dispensing system **1344** includes a dispensing device that can be used to fully wash the equipment placed into the container and stored within the enclosure **1306**. In an embodiment, multiple wash cycles (e.g., a sterilization cycle, a rinse cycle, etc.) may be performed on the effectors and possibly other components.

(115) The dispensing system **1344** may include an interchangeable fluid reservoir **1346** (see FIG. **13**) that stores liquids (e.g., sterilizing agents, heat absorbing agents, lubricating agents, etc.) that are used with the therapy system **1300** and from which one or more pumps supply fluid to the dispensing device and/or nozzle(s) of the therapy device(s). The dispensing system **1344** may also be configured to pre-treat surfaces of the effectors and/or therapy devices with lubricants (e.g., oils) or other liquids. The reservoir **1346** may be detachably coupled to the enclosure **1306** and may be replaced when the liquid supply drops below threshold levels. The sterilization system **1342** may further include fans, blowers, and/or heating elements to dry off the tissue treatment devices **1304**

at the end of a wash cycle. In other embodiments, the enclosure **1306** may define a second cavity—separate from the storage area of the interior cavity **1308**—that is used to sterilize the tissue treatment devices **1304** and other add-on equipment after use. The sterilization system **1342** may be affixed to an inner surface **1348** or structure within the interior cavity **1308**. The sterilization system **1342** may have a door positioned on top and/or side of an internal container and/or the enclosure **1306**, and enable an operator to place effector(s) within the sterilization system **1342**. As previously described, the sterilization system **1342** may include fluid that may be sprayed onto the effector(s) and/or other components placed therein, and automatically perform a sterilization cycle (e.g., spray/soak, rinse, rinse, dry, illuminate with UV lights to disinfect the effort(s)).

(116) Returning to FIG. **1**, the computing system **103** (e.g., computing device, etc.) may be configured to monitor and manage treatment sessions of patients by an operator of the therapy system **100**, and/or may include a control system configured to control operation of the fascia tissue treatment device(s) **104**. The computing system **103** may include a controller having a processor and memory storing software for the system for supporting patient treatments (e.g., different treatment regimens, patient data received from sensors, such as pressure sensor(s), speed sensor, torque sensor, temperature sensor(s), etc., onboard the treatment device **104** and/or the user interface, etc.). The computing system **103** may include a user interface **140** coupled to the base station **102**, to receive user commands and display treatment and/or operating information for the therapy system **100**. For example, the computing system **103** may include a tablet (e.g., iPad, etc.) and/or another input/output that is detachably coupled to the base station **102**.

(117) As further shown in FIG. **1**, the tablet may be pivotably and/or rotatably coupled to the base station **102** via a mounting stand **142** that allows the user to reposition the tablet (e.g., a display screen) with respect to the base station **102**. In other embodiments, as shown in FIGS. **14**A-**14**B, a user interface **1404** (e.g., a tablet) may be mounted on a swivel **1444** along the upper wall **1418** of the enclosure **1406**. The swivel **1444** may allow 360° rotation of the user interface **1404** with respect to the base station **1402**. The swivel **1444** may also include a receiver **1446** that engages the user interface **1404** and is structured to tilt relative to the upper wall **1418** of the enclosure **1406** (FIG. **17**). The user interface **1404** may be remotely connected (e.g., via Bluetooth, WiFi, etc.) and/or hardwired to the base station **1402** (e.g., via the mounting stand, etc.). In some embodiments, as shown in FIG. **1**, the user interface may also include buttons, toggles, and/or another form of user input device **144** disposed on the base station **102** (e.g., enclosure **106**) and configured to control certain functionality of the therapy system **100**. For example, the user interface **140** may be configured to enable an operator to set certain conditions (e.g., shut off, alarms, alerts, self-cleaning, etc.) on the base station **102** to perform certain functions, such as allowing a user to be notified of certain conditions (e.g., a temperature of power generator being too high, a temperature of the treatment device **104** being too high, a torque of the treatment device **104** being too high, etc.).

(118) In one embodiment, the user interface **140** may be configured to provide treatment guidance to an operator, such as modalities to perform using specific treatment devices **104** and/or effectors for certain durations and at different speeds. The operator may perform the treatment, and the user interface **140** may automatically collect treatment data (e.g., duration of procedure, speeds of effectors, treatment modality, etc.). In one embodiment, the computing system **103** is configured to (i) curate operator sign-in, scheduling of treatment, and collection of patient data (e.g., treatment notes from a clinician, treatment start and end times, etc.), and (ii) guide an operator in applying the treatment device **104** and pre-treatment procedures. The computing system **103** may include interactive software that allows a user to obtain additional information regarding one or more treatment operations (e.g., how to activate a treatment device **104**, how to manipulate the treatment device **104** during treatment, etc.). The computing system **103** may also provide video demonstrations to further clarify operation of different aspects of the therapy system **100**, as will be further described.

(119) In another embodiment, the computing system **103** is configured to automatically or semi-automatically (e.g., in combination with inputs from a clinician) analyze inputs to develop treatment plans for the patient. The computing system **103** may be configured to define a therapy regimen (e.g., a list of treatments to be performed), and compare pre-treatment and post-treatment results (e.g., historical results over the treatment period), among other automated and non-automated features. In this embodiment, the computing system **103** is also configured to control a treatment device operation to administer treatment and/or to provide feedback to the user and/or patient based on sensor data, as previously described and as will be further described herein.
(120) The design of the therapy system described with reference to FIGS. **1-2** and FIGS. **3**A-**3**C should not be considered limiting. Many alternatives and combinations are possible without departing from the inventive principles disclosed herein. For example, FIGS. **15**A-**15**B, FIGS. **16**A-**16**B, and FIGS. **17-20** show various other illustrative embodiments of a base station for a therapy system that include different designs for the enclosure, user interface, access panel, and/or mounts for the fascia tissue treatment devices. For example, FIGS. **15**A-**15**B show a therapy system **1500** having an access door **1524** that pivots away from an enclosure **1506** of the therapy system **1500**. FIGS. **16**A-**16**B show a therapy system **1600** having an enclosure **1606** having a profile that is shaped to match the natural contours of a human body (e.g., a frustoconical shape with curved sidewalls). FIGS. **17-20** show various other shapes for an enclosure **1706**, **1806**, **1906**, **2006** of a treatment system **1700**, **1800**, **1900**, **2000**.
(121) Referring to FIGS. **21-22**, FIGS. **23**A-**23**D, and FIGS. **24**A-**24**B various illustrative fascia tissue treatment devices **2100**, **2200**, **2300**, **2400** (e.g., therapy devices, etc.) are shown for use with the fascia tissue therapy system. The treatment devices **2100**, **2200**, **2300**, **2400** (e.g., heads, etc.) are configured to apply a treatment to a user by moving a tissue treatment assembly (e.g., an effector, a tissue treatment head, etc.) along a user's skin. As shown in FIGS. **23**A-**23**B, the treatment device **2300** includes a housing **2302**, a tissue treatment assembly **2304** coupled to the housing **2302**, and an actuator **2306** configured to power the tissue treatment assembly **2304**. The treatment device **2304** may also include at least one sensor **2308** (e.g., pressure, speed, torque, temperature, etc.), a heating element (e.g., IR lighting), a user interface, and/or other electronic equipment.
(122) FIGS. **25**A-**25**B show exploded views of an illustrative fascia tissue treatment device **2500** that is similar to the devices of FIGS. **21-22**, FIGS. **23**A-**23**D, and FIGS. **24**A-**24**B. As shown in FIG. **25**A, the treatment device **2500** is detachably coupled to the tether **2510** of a therapy system that electrically couples the actuator and other onboard electronics to the therapy system. The treatment device **2500** is also detachably coupled to the tissue treatment assembly **2504** and at least one handle **2512** for the treatment device **2500**. In this way, different tissue treatment assembly designs may be used with the same treatment device **2500**. The treatment device **2500** may include a quick-connect interface **2514**, such as a chuck, snap-fit connector, and/or another form of detachable coupling to facilitate removal and replacement of the tissue treatment assembly **2504**. The tissue treatment assembly **2504** may couple to a mounting member **2516** (e.g., a center mounting member disposed centrally along an end of a housing **2502** of the treatment device **2500**, etc.) of the tissue treatment assembly **2504**. For example, the tissue treatment assembly **2504** may include a support member **2518** disposed at a central position on the tissue treatment assembly **2504** (e.g., a panel **2520** of the tissue treatment assembly **2504**) that extends axially away from the tissue treatment assembly **2504** (e.g., the panel **2520**). In other embodiments, the support member **2518** may be coupled to or otherwise engage the tissue treatment assembly **2504** at an intermediate radial position approximately half-way between a central axis **2522** of the tissue treatment assembly **2504** and an outer perimeter **2524** of the tissue treatment assembly **2504**. In yet other embodiments, the treatment device **2500** may be configured to engage the tissue treatment assembly **2504** along an outer perimeter **2524** of the tissue treatment assembly **2504**. Still yet, the tissue treatment assembly **2504** may be centrally connected to a motor shaft of the actuator.

Engaging the tissue treatment assembly **2504** along an intermediate or outer radial position may reduce torque on the actuator during treatment (due to the smaller moment arm of the tissue treatment assembly **2504**), reduce deflection of the tissue treatment assembly **2504** under an applied axial force, and ensure more uniform engagement between the treatment elements of the tissue treatment assembly **2504** and a user's fascia tissue.

(123) The treatment device **2500** may be structured for use both with and without the handle **2512** depending if the treatment device **2500** is to be controlled by an operator using one or two hands. The configuration of the treatment device **2500** may, in part, depend on modalities to be performed by the particular treatment device **2500** (e.g., leg and back treatment devices are larger than face and feet treatment devices), weight of the treatment device **2500**, speed of the motor, diameter of the tissue treatment assembly **2504**, and so on. For example, FIGS. **26**A-**26**B show an example treatment device **2600** that is configured for two-handed operation. As shown, a housing **2602** of the treatment device **2600** includes a first end portion **2626** (e.g., a lower portion, etc.) that is structured to engage with a tissue treatment assembly, a second end portion **2628** (e.g., an upper portion, etc.) spaced axially apart from the first end portion **2626** that is detachably coupled to the tether **2610**, and an intermediate portion **2630** (e.g., a central portion, a middle portion, etc.) that extends between the first end portion **2626** and the second end portion **2628**. The intermediate portion **2630** may have a smaller cross-section than the first end portion **2626** and the second end portion **2628** such that the first end portion **2626**, the second end portion **2628**, and the intermediate portion **2630** together form an hourglass shape. As shown in FIG. **26**B, the intermediate portion **2630** of the housing **2602** is shaped so that an operator may wrap one of his or her hands around the intermediate portion **2630**, which can be used to guide the treatment device **2600** across a patient's skin. As shown in FIG. **25**B, the housing **2502** may include pads **2532** that enhance a user's grip on the treatment device **2500**. The pads **2532** and/or the housing **2502** may be at least partially formed from a soft silicone material to provide a pleasing velvety feel to the user when manipulating the treatment device **2500**. As shown in FIG. **26**A, the treatment device **2600** may include a locking mechanism **2634** (e.g., “dead man” switch) that allows the operator to turn the device ON until the locking mechanism **2634** is released. Of course, the treatment device **2600** may include a number of sensors, such as timer, impact switch, timer, temperature, etc., that, if any of the sensors detects a problem (e.g., treatment device is dropped), then the treatment device **2600** may automatically be turned OFF.

(124) As shown in FIG. **26**A, the therapy device may also include a user interface **2635** configured to control operation of the actuator and tissue treatment assembly **2604**. In one embodiment, the treatment device **2600** may have a switch **2636** (e.g., a trigger) that causes an actuator (e.g., motor) to turn ON. Alternatively, or in combination, a speed of the actuator(s) may be controlled by the switch **2636** that duals as a speed control. Alternatively, a speed controller (e.g., knob or selector on the treatment device **2600**) on the treatment device **2600** may enable the operator to change speed of the actuator. If the user interface **2635** operates to control the therapy or treatment device **2600**, then speed control may be established on the user interface **2635**. Still yet, if the treatment is programmed into the user interface **2635**, then speed profiles for the treatment may be established thereon so that the operator can use the pre-programmed speed or may override.

(125) In an embodiment, the switch **2636** (e.g., trigger, button, etc.) is disposed along the intermediate portion **2630** that controls activation of the actuator. In other embodiments, the switch **2636** may also be structured to allow a user to control the speed and/or other functionality of the treatment device **2600**. For example, the user interface **2635** on the treatment device **2600** and/or base station may also include a setting selector that allows a user to manually adjust operational limits, effector stiffness, and/or other control parameters for the treatment device **2600**. In some embodiments, the treatment device **2600** may also include an auto shut-off switch and/or “dead man's” switch (e.g., the locking mechanism **2634**) that deactivates and/or locks the actuator in the event a rapid shut down is required. In some embodiments, the dead man's switch may also be

connected to pressure/force, and/or torque measurement sensors within the treatment device **2600** and/or tissue treatment assembly to automatically shut down the system when certain threshold values are satisfied (e.g., when the pressure and/or torque exceeds threshold values).

(126) In some embodiments, the user interface **2635** may also be configured to provide a curated treatment procedure (e.g., modality guidance, etc.) to facilitate user interaction with the treatment device **2600** and to improve treatment effectiveness. For example, the user interface **2635** may provide step-by-step instructions informing the user of where to place the treatment device **2600** and how to hold and manipulate the treatment device **2600** during treatment. For example, the treatment device **2600** may be configured to present a video to a user (e.g., clinician, etc.) to inform the user of each step in the treatment plan, which devices to use, how to use each device. FIG. **27** shows a storyboard **2700** for an example video demonstration of a treatment plan. The demonstration includes informing the user of how to manipulate and operate base station equipment, which therapy devices to use and in what order, and the specific treatment steps to be followed for each device. As shown, the storyboard provides for pre-treatment of fascia tissue, treatment of the fascia tissue, and post-treatment. It should be understood that more specific detail showing differences for different modalities may be provided. For example, pre-treatment of a leg or torso with more muscle mass or fat content may be different than pre-treatment of a face or scalp with less muscle mass and fat content. A treatment plan may start with fascia tissue that is closer to the skin and then deeper tissue treatments may occur over time to gradually work towards restructuring fascia tissue that is deeper. Imaging (e.g., ultrasound imaging) may be used as “feedback” to an operator and/or the system, if so configured, to see how treatments are progressively restructuring fascia tissue starting from the surface and working deeper. Such treatment plans may, of course, be tailored to specific fascia tissue

(127) As shown in FIG. **26**B, the second end portion **2628** is shaped so that a user may apply a downward force onto the tissue treatment assembly **2604** and toward a patient's skin, thereby causing the tissue treatment assembly **2604** to rotate while the pressure is being applied during treatment. By applying more pressure, the treatment to fascia tissue that is deeper (e.g., towards an underlying leg bone). The second end portion **2628** may define a planar upper surface that extends at an oblique angle relative to the central axis of the housing **2602**. A user may engage their palm with the upper surface to apply a downward force toward the skin during treatment. The angled upper surface may improve ergonomics of the treatment device **2600** and allow a user to maintain pressure on the tissue over a longer period of time before becoming tired. It will be appreciated that the shape of the housing **2602** may be different in other embodiments.

(128) In an embodiment, one or more feedback mechanisms may be included in the treatment device **2600** that enables feedback to be given to the operator. For example, lights for illumination, speaker for audible, vibration element for tactile feedback may be provided, where the feedback may be feedback for temperature (of the motor), high current draw (representing high torque (e.g., using current sensor)), treatment cycle start/end (using clock or motion sensor), force being applied is too low or too high (using one or more pressure sensor(s), etc.).

(129) Referring to FIG. **28**, another illustrative treatment device **2800** (e.g., therapy device, etc.) is shown that includes a removable handle **2812**. The removable handle **2812** may be removable from handles of a structural holster or strap **2813**, where the handles and/or strap **2813** may be formed of or include silicone rubber (or any other material). The handle **2812** may be used when using the treatment device **2800** to treat elongated body areas such as a patient's legs and back, while reducing stress to a user's hands in applying force to the treatment device **2800**. The handles and/or a strap **2813** may engage an end portion (e.g., a lower portion, etc.) of the housing of the treatment device **2800**, proximate to the tissue treatment assembly, or any other location of the housing. As shown in FIG. **29**, the handle **2812** may be formed with the strap **2813** or holster that engages with a second end portion of the housing to couple the handle **2812** to the housing. In other embodiments (as shown in FIGS. **37-40**), the handle **2812** may engage with another portion of the

housing away from the tissue treatment assembly (e.g., the intermediate portion of the housing, a first end portion of the housing opposite from the second end portion, etc.). The strap **2813** may mount to the treatment device **2800** via a snap-fit connection, a fastener, and/or other detachably coupling. As shown in FIG. **29**, the handle **2812** also includes two posts **2815** that are coupled to the strap **2813** and extend radially away from the strap **2813**. The posts **2815** may be substantially cylindrical rods made from silicone rubber or another soft, yet suitably stiff material that a user can interact with to manipulate the position of the treatment device **2800** and force applied to the skin during treatment. As shown in FIG. **30** and FIG. **31**, the handles may be separated from the treatment device **2800** depending on the desired treatment application, and/or to facilitate treatment of curved body areas such as the knee, neck, and others.

(130) Referring again to FIGS. **28-29**, the handles **2812** and strap **2813** may be used during treatment and/or storage of the treatment device **2800**. In other words, the handles **2812** and the strap **2813** may be held by an operator treating a patient or may simply be used for storage of the treatment device **2800**. If used during treatment, having two handles **2812** enables the operator to apply pressure equally, if desired, to both handles **2812**, thereby applying force by the tissue treatment assembly to the patient in an equal manner. The strap **2813** may be configured to engage with the treatment device **2800** in a manner that still allows for a user interface (e.g., electronic display, LEDs, audio, etc.) disposed thereon to provide treatment and/or feedback information to the operator.

(131) FIGS. **32-35** show various alternative illustrative handle designs for the treatment device **2800** of FIGS. **30-31**. As shown in FIGS. **32-33**, the handle **3212**, **3312** may include a strap **3213**, **3313** that circumscribes the housing of the therapy device. In other embodiments (e.g., as shown in FIGS. **34** and **35**), the handle **3412**, **3512** wraps only partially around the treatment device and includes a slot to facilitate assembly of the handle **3412**, **3512** to the treatment device (e.g., by using a flexible strap **3413**, **3513** that allows increasing the size of the opening at the slot during assembly to clamp the flexible strap onto the housing). It should be understood that while the handles and/or strap may be flexible, alternative embodiments may have the handles and/or strap be rigid (i.e., not easily bent). Rather than the handles and strap being added, a housing of the therapy device may have handles formed therewith and the handles may be monolithic with the material of the housing of the therapy device.

(132) The design of the housing for the treatment device may be different in various embodiments. For example, FIGS. **36-37** show an illustrative treatment device **3600** that includes a housing having a lower, disk-shaped portion **3602** and an upper, conically-shaped portion **3604** that extends axially away from an upper surface of the lower, disk-shaped portion **3602**. As shown in FIG. **36**, a user may press against an upper surface of the disk-shaped portion **3602** to facilitate treatment of large body areas such as a patient's back or legs. As shown in FIG. **37**, the user may engage with the conically-shaped portion **3604** for finer control and movement of the treatment device **3600**.

(133) FIG. **38** shows yet another illustrative treatment device **3800** (e.g., a therapy device, etc.). The treatment device **3800** includes a housing design that is similar to the housing of the treatment device **3600** of FIGS. **36-37**, but that has a disk-shaped portion **3802** with a smaller cross-section to facilitate single-handed manipulation or use for treatment of body areas where increased maneuverability of the treatment device **3800** is beneficial to the operator.

(134) FIG. **39** shows yet another illustrative tissue treatment device **3900** (e.g., a therapy device, etc.). The treatment device **3900** includes a saddle-shaped housing **3902** defining a saddle **3930** that is designed to allow an operator to wrap his or her hand around the top of the treatment device **3900** to both apply vertical and/or angular force and manipulate a position of the treatment device **3900**. The housing **3902** may be shaped such that a central axis **3922** of the housing **3902** is approximately aligned with a user's forearm when the operator's hand is wrapped around the saddle **3930** which, in one embodiment, centers any downward force applied to the housing **3902** during use. The saddle **3930** is also shaped to accommodate the natural angle of a user's wrist to reduce

strain on the wrist during use.

(135) As shown in FIG. **39**, the treatment device **3900** may include various add-ons to facilitate monitoring and diagnoses of damaged and/or abnormal fascia tissue, improve manipulation and restoration of fascia tissue, and to reduce the risk of damage to the treatment device **3900** during operation. For example, the treatment device **3900** in FIG. **39** may include a cooling system **3932** that is structured to cool the actuator and/or other electronic components contained within the housing **3902** during treatment. The cooling system **3932** may include a fan and/or blower disposed within the housing **3902** that forces air across the actuator to exit through an exhaust port in the housing **3932** (shown as rear exhaust port **3934** in FIG. **44**A). In other embodiments, the treatment device **3900** may include another form of cooling (e.g., liquid cooling) to cool internal components. Hydraulic or other cooling techniques may also be possible. As previously described, a direct drive motor may be utilized to reduce the amount of heat produced by the actuator/motor.

(136) The treatment device **3900** may also include at least one sensor **3936** (e.g., force sensor configured to generate sensor data indicative of a force applied to the sensor, etc.) for monitoring device operations and a transceiver **3938** (e.g., WiFi, Bluetooth, etc.) for communicating sensor data with the controller and/or other networked systems. For example, the treatment device **3900** may include at least one pressure sensor structured to measure a force applied to portions of the tissue treatment assembly (e.g., the effector, etc.), between the tissue treatment assembly and a patient's skin. In at least one embodiment, the treatment device **3900** includes multiple pressure sensors located in different areas or quadrants of the tissue treatment assembly (e.g., at different circumferential and/or axial positions along a panel (e.g., an effector panel, etc.) of the tissue treatment assembly as shown in FIG. **39**, at equal intervals across the panel, at a location of each tissue treatment element along the panel, etc.).

(137) The pressure/force sensors may be configured to provide feedback to the user via a user interface of the computing device and/or treatment device **3900** to indicate the measured pressure at the location of each pressure sensor (e.g., via a user interface of the treatment device **3900**, or via the user interface on the base station). For example, the treatment device **3900** in FIG. **39** may include multiple indicator lights **3940** disposed along an outer perimeter **3942** of the housing **3902** and facing radially away from or vertical from the housing **3902**. In other embodiments, the location of the indicator lights **3940** on the treatment device **3900** may be different. For example, the lights may be disposed on an upper wall of the treatment device, as shown in FIG. **37** (e.g., indicator light **3640**). The indicator lights **3940** may be configured to provide an indication of the actual force and/or pressure being applied to the tissue treatment assembly (or a portion of a panel of the tissue treatment assembly) relative to a target force and/or pressure for treatment. For example, as shown in FIG. **39**, the treatment device **3900** may include a plurality of sensors **3936** configured to measure a force applied to quadrants **3944** of a panel **3920** of the tissue treatment assembly.

(138) A controller (e.g., at the transceiver **3938**, etc.) may be communicably coupled to the sensors **3936** and may be configured to determine forces being applied to the quadrants **3944** of the panel **3920** while the actuator is in an ON state based on sensor data from the sensors **3936**. The indicator lights **3940** in a first circumferential quadrant of the treatment device **3900** may indicate whether the pressure and/or force applied in a nearest quadrant **3944** of the panel **3920** is less than a target pressure and/or force based on a color of the lights (e.g., red, yellow, green, blue, etc.), an amount of flashing, and/or other visual indication. In an embodiment, if a target range of forces (i.e., low force threshold and high force threshold) is established for a particular modality or treatment, then in the event that force (e.g., force in a quadrant of the effector) is sensed to be too low, then lights (e.g., LEDs) may be illuminated to be one color (e.g., blue) and if sensed to be too high, then lights may be illuminated to be in another color (e.g., red). If the pressure is sensed to be within range of the desired treatment levels, then another color (e.g., green) may be displayed by the indicator lights **3940**. If sensors **3936** configured in quadrants **3944** are used, then the feedback (e.g., lights)

may be displayed in relation to each pressure measured in the quadrants **3944** associated with the sensors **3936** in the respective quadrants **3944**. Other color schemes may be utilized, as well. Moreover, the pressure sensor feedback signals may be collected by the user interface and stored thereon, thereby enabling the user interface or another computing device to correlate pressure, therapy devices, operator, etc. with fascia tissue restructuring progress, for example.
(139) The treatment device **3900** may also include a speaker to provide audible and/or tactile notification of the applied vs. desired level of pressure and/or force to a user. In some embodiments, the sensors **3936** (e.g., force and/or pressure sensors, etc.) may also be used to determine an average pressure across the tissue treatment assembly (e.g., by averaging measurements from each sensor **3936**), a peak pressure across the tissue treatment assembly, and/or to facilitate calibration of the treatment device **3900** (e.g., to facilitate adjustment of an angle of the tissue treatment assembly relative to the housing **3902** of the treatment device **3900** after a new tissue treatment assembly has been installed). If, for example, pressure is too low, a change of an audible signal (e.g., Geiger counter tones, frequency, pitch, and/or volume of an audible signal, etc.) or a notification signal (e.g., tone, beep, etc.) may be produced.
(140) The tissue treatment device **3900** may also include other types of sensors and/or transducers to facilitate assessment of fascia tissue abnormalities prior to, or during, treatment. Examples of fascia tissue layers **4000** and damage is shown in FIG. **40**A. Outer and inner structural fascia tissue layers (thin layer beneath the top skin surface layer) are shown to separate skin (top surface layer), subcutaneous fat (beneath the top outer structural fascia tissue layer), and muscle (layer below the subcutaneous fat later) layers. Also shown are inter-structural fascia tissue fibers that extend between the structural tissue layers. Problem areas occur in regions of high fascia tissue density of the inter-structural fascia tissue fibers, which may also be described as adhesions of the fascia tissue. As shown, pockets/dimples **4002** (commonly referred to as cellulite) are formed on the skin surface as a result of the adhesions of the inter-structural fascia tissue fibers. While skin aesthetics may result from adhesions of the fascia tissue, more health-related problems may also result. Such health-related issue may include pinched nerves, reduced blood circulation, muscle pain due to the inter-structural fascia tissue fibers having reduced flexibility, tearing of the fascia tissue if the fascia tissue fibers are too intertwined, and so on.
(141) FIG. **40**B is an illustration of a cross-sectional view of an illustrative leg **4006** showing illustrative structural and inter-structural fascia tissue and muscle, including anterior or quadriceps muscles, posterior or hamstring muscles, and medial or adductor muscles. The human body is formed of a “web” of interconnected systems. When one part of the system slacks, the rest may suffer. Nowhere is that more true than with fascia tissue. This web of fascia tissue impacts every bone, organ, muscle, and bit of soft tissue in the body. Healthy or unhealthy fascia tissue may have a big impact in quality of life of an individual. For example, unhealthy fascia tissue may mean that fascia tissue has adhesions such that the fascia tissue may squeeze a muscle to restrict blood flow or circulation to an extremity or pinch a nerve to cause inflammation or pain. As shown, inter-structural fascia tissue form sheaths or barriers around and between the different muscles of the leg. Similar structural and inter-structural fascia tissues are located through the entire human (and animal) body.
(142) The human body generally has four types of fascia tissue, each with its own role to play. The fascia tissue types include structural fascia, inter-structural fascia, visceral fascia, and the spinal straw. The two most commonly known types of fascia tissue are traditionally called “superficial fascia” and “deep fascia,” which are both part of the structural fascia. These terms can be somewhat misleading because superficial and deep fascia tissue barely begin to describe the “big picture” of fascia tissue and the functions the fascia tissue plays for the human body. In fact, if a cadaver dissection is examined, you fascia is not stacked in neat, linear layers, but are rather form an interwoven, consecutive piece of fascia tissue, like a 3D spider web.
(143) More specifically, fascia is a system of internal connective tissue that forms fibrous bands or

sheets throughout the body. The structural fascia, including superficial and deep fascia, surrounds muscles, nerves, blood vessels, and internal organs. Fascia tissue is often considered the body's first line of defense against infection and other pathogens, and the fascia tissue creates an environment for tissue repair to occur.

(144) Fascia tissue may be thought of as a layer of plastic wrap holding the contents of a sandwich together, but in this case, the sandwich is the body including organs and tissue (e.g., muscle, bone, etc.) therein. The top layer of the wrap is the superficial fascia or structural fascia tissue and the bottom layer is deep fascia, which includes inter-structural fascia (found within internal structures, such as muscles), visceral fascia (“goopy” fascia, found in the abdomen), and spinal straw (found within the vertebral column and all around the spine).

(145) While fascia tissue is essentially a single interconnected piece of tissue, the fascia tissue does look and operate differently depending on where the fascia tissue located. For example, the superficial fascia tissue (located just under the skin in the subcutaneous tissue) is filled with sensory receptors and functions to anchor skin to the tissues and organs below. The superficial fascia tissue is a thin, fibrous material that is classified as loose connective tissue throughout the skeletal muscles. Superficial fascia is a subcategory of the structural fascia.

(146) Deep fascia tissue is a little different from superficial fascia tissue as the deep tissue is found further below the surface and surrounds, separates, and supports every structure in the body. Deep fascia tissue is also filled with sensory receptors, and is sensitive to changes in pressure and movement—especially through massage and stretching. Deep fascia tissue is also a subcategory of the structural fascia.

(147) As a summary, superficial fascia tissue and deep fascia tissue have more similarities than differences, where both are made of connective tissue and contain bundles of collagen and elastin fibers. The primary difference between superficial fascia tissue and deep fascia tissue is that superficial fascia tissue is composed of loose connective tissue and is found in the layer between the dermis layer of your skin and muscles, while deep fascia tissue contains dense connective tissue and is found within and between your muscles, where the muscles and fascia tissue work together in the extracellular matrix. Researchers used to think of deep fascia tissue as a second layer of superficial fascia tissue, but have recently come to understand that the superficial fascia tissue extends far deeper in the body than previously credited. Treatment of the superficial fascia tissue and deep fascia tissue utilizing the fascia tissue treatment devices, as described herein, allows for “fascial shearing” and restructuring of the different types of fascia tissue, thereby resulting in healthy fascia tissue and alleviation or elimination of effects resulting from the fascia tissue having adhesions or otherwise damaged, as shown in the ultrasound images of FIGS. **41-44**. Utilizing the fascia tissue treatment devices with effectors and treatment processes described herein, it is possible to treat both the superficial fascia tissue and deep fascia tissue in a manner that provides for “fascial shearing” and is able to remove adhesions of the fascia tissue at any depth of the body. Treating deep tissue, of course, may take longer than treating superficial fascia tissue.

(148) In various embodiments, the treatment device may include an ultrasound system and/or sonography imaging device that uses high-frequency sound waves to produce images of the fascia tissue, including different layers of fascia tissue depending on the ultrasound sensor and imaging software. The ultrasound system may form part of the computing device for the therapy device and may be configured to transmit images to the controller for further analysis. The controller may be configured to utilize the images to identify problem areas along a patient's body. In an embodiment, a medical professional and/or operator may be create a treatment plan based on the imaging of the fascia tissue with the assistance of the user interface. In an alternative embodiment, the user interface may be configured to determine a treatment plan based on the images, and evaluate treatment progress over time based on repeated images of the same area, as further described hereinbelow.

(149) For example, in a pre-treatment mode, the controller may be configured to guide a user to

place the therapy device at different locations along a patient's body to map various areas of the body (see ultrasound images **4100**, **4200**, **4300**, **4400** in FIGS. **41-44**). The controller may be configured to operate the ultrasound system at each location to obtain images across a range of resolutions and/or tissue depths (e.g., three tissue depths at each location, etc.). The controller may then analyze the images to identify areas of high fascia tissue density, fragmentation, and/or other image characteristics that are indicative of fascia tissue abnormality. The controller may be configured to perform image analysis based on predetermined algorithms in memory, and/or by transmitting the images to a remote computing system (e.g., a cloud computing system) for further analysis. In at least one embodiment, the controller may be configured to use images from different patient screenings (in combination with user inputs identifying problem areas) as a training set for machine learning and/or artificial intelligence (AI) algorithm that can automatically identify fascia tissue damage and/or abnormalities. The machine learning algorithm may also be used—in combination with historical data from patient treatment protocols—to determine a treatment plan based on the detected fascia tissue abnormalities. The machine learning algorithm may utilize a k-nearest neighbor (KNN) algorithm and/or another classification algorithm to classify fascia tissue characteristics relative to common fascia tissue characteristics of other patients so that successful treatments that work for different types of fascia tissue characteristics can be applied to a patient. The AI identification of certain fascia tissue characteristics may be specific to specific body areas (e.g., lower back, calves, scalp, etc.) requiring treatment, the type of treatment that is required, and/or the required number of treatments. The controller may be configured to utilize the ultrasound images to determine or select particular effector characteristics (e.g., size, shape, finger orientation, etc.) and control parameters (e.g., operating speed, force, etc.) to use to improve treatment efficacy. Referring to FIG. **45**, an illustrative method **4500** of fascia tissue analysis is shown. The method **4500** includes transmitting a scan command (at operation **4502**) to a treatment device to perform a fascia tissue scan (e.g., via ultrasound, etc.), receiving scan data (at operation **4504**) (e.g., images of the tissue at different depths, etc.), correlating scan data with previously scanned tissue (at operation **4506**), identifying changes in scan data (at operation **4508**) (e.g., by comparing scan data with images from previous scans, etc.). In some embodiments, the method **4500** also includes determining a treatment plan based on scan data and/or identified changes in scan data (at operation **4510**). In other embodiments, the method **4500** may include additional, fewer, and/or different operations.
(150) The tissue treatment device (e.g., the therapy device, fitness device, etc.) may also include sensors that interface with the computing device to ensure the correct device and/or tissue treatment assembly is being used for a prescribed treatment protocol and to monitor operation of the treatment device during treatment (e.g., the average applied force, torque, rotational speed, duration of use, etc.). For example, each treatment device may be coded with a type, size, etc., and each tissue treatment assembly (e.g., each effector, etc.) may also be coded with a type, size, finger or treatment element type, shape, etc. In coding the treatment devices and tissue treatment assemblies, chips, such as RFID chips or processor chips or memory chips, may be coded with identifying information, thereby enabling the user interface to receive and record the identification information so that historical records may be maintained in an automatic manner. The controller may record this data at periodic intervals (as shown in FIGS. **44**C-**44**F), which can be fed back into the machine learning algorithm to update and/or improve treatment prescription.
(151) The treatment device may also include other forms of skin treatment devices. For example, the treatment device may include an infrared element to treat a patient's skin and/or underlying tissue, a radio frequency skin tightening system, and/or ultrasonic tissue treatment system. In some embodiments, the treatment device also includes a lubrication system (e.g., nozzles, a micropump, etc.) that is structured to dispense a lubricating agent to a patient's tissue to reduce friction between the effector and the patient's skin during rotation of the effector, thereby enabling the operator to continuously treat the patient without having to stop using the therapy device. In an embodiment,

feedback from a torque sensor may be utilized to determine when additional lubricant should be applied to the skin, and the lubrication system may automatically dispense or be caused to dispense lubricant to the skin by the therapy device. These auxiliary treatment devices may be at least partially contained within the housing of the therapy device and/or may be attached to the treatment device in place of, or in addition to, the tissue treatment assembly. In yet other embodiments, these auxiliary treatment devices may form their own treatment device that may be detachably coupled to the base station. The lubrication may be stored in a reservoir stored in the enclosure that may be refillable or cartridges of lubrication may be replaced in the lubrication system, as previously described.

(152) The actuator of the treatment device is configured to power rotation of the tissue treatment assembly. The actuator may include a direct drive motor that engages the tissue treatment assembly without any intervening transmission or gear set. Among other benefits, using a direct drive motor reduces heat produced by the treatment device and wear on the internal components. The direct drive motor may also reduce the weight of the treatment device due to the increased torque to weight ratio of the motor. In other embodiments, the actuator includes another form of electromechanical device (e.g., brushless direct current motor, electromagnetic device, etc.). In other embodiments, the actuator includes a pneumatic motor that is configured to drive rotation of the tissue treatment assembly using air pressure. For example, the pneumatic motor may consist of a turbine in the treatment device housing that is driven into rotation by compressed air provided through the tether to the base station. In other embodiments, the actuator may include a hydraulic motor and/or system.

(153) In some embodiments, the actuator may be configured to move the effector and/or individual treatment elements along a single direction (e.g., linearly back and forth, etc.) instead of, or in addition to, a rotational direction. For example, FIG. **46** shows an illustrative fascia tissue treatment device **4600** in which the actuator is structured to move individual treatment elements (e.g., claws, etc.) back and forth along a single direction **4602**. The elements may be staggered at different positions along each actuator or aligned. FIG. **47** shows another illustrative fascia tissue treatment device **4700** in which the actuator includes a radial bar that connects multiple tissue treatment elements and alternates direction (e.g., rotational direction) to move each element along a semi-circular path. In yet other embodiments, the treatment device may include a shaft with offset lobes (e.g., similar to an engine crankshaft) to move the treatment elements along guides (e.g., slots) in the housing. In yet other embodiments, the treatment device may include feet (e.g., made from silicone and/or coated with silicone or another suitable material) that may translate back and forth along the treatment device, as shown in FIGS. **48-54**. For example, FIGS. **48-50** show a treatment device **4800** that includes tissue treatment assembly including a plurality of feet **4804** (e.g., pads, panels, etc.) that are coupled to the treatment device **4800** separately from one another. The feet **4804** move in alternating directions along a linear path. The feet **4804** may move in unison with one another or at least partially in opposition depending on the treatment modality. FIGS. **51-54** show a treatment device **5100** that includes only two feet **5104** that move in opposing directions when the actuator is in an ON position. It should be understood that the size of the feet and range of motion of the feet may be different in various embodiments.

(154) In yet other embodiments, the actuator may be another form of rotational and/or vibrational device (e.g., an electromagnetic device, a rotating cable extending through the tether and driven by a motor in the base station, etc.). Vibration of the treatment device may be random or non-random. For example, the device may rotate and vibrate up-and-down. The device may alternatively rotate and randomly vibrate. For example, FIG. **55** is an isometric view of an illustrative treatment device **5500** that includes a linear actuator **5506** that is configured to vibrate or otherwise translate a treatment element **5508** back and forth along a linear direction (e.g., parallel to a central axis of a support bar to which the treatment elements **5508** are mounted, etc.). The treatment element **5508** (e.g., the actuator **5506**) may thereby thump, hammer, or pulsate against the patient's tissue during

operation.
(155) In some embodiments, the actuator **5506** may also be configured to vibrate the tissue treatment assembly (e.g., the effector, etc.) in combination of spinning or another form of movement.
(156) In some embodiments, the treatment device **5500** (e.g., the actuator **5506** of the treatment device **5500**, the tissue treatment assembly **5504**, etc.) may be configured to apply micro-currents or certain electrical currents through the tissue treatment assembly **5504** and/or another treatment accessory to electrically stimulate the tissue in combination with spinning and/or vibration and/or to apply light in certain wavelengths to enhance fascia tissue treatment by the treatment device **5500**. For example, as shown in FIG. **55**, the treatment device **5500** may include a treatment accessory **5540** that is detachably coupled to the treatment device **5500** that is configured to apply electrical currents and light in certain wavelengths to a patient's skin. The treatment accessory **5540** may include brushes **5542** arranged in rows along the treatment accessory **5540**, although various other arrangements may be used in other embodiments. The bristles of the brush **5542** may be made of any suitable material including nylon, polypropylene, horse hair, feather, or any other natural or synthetic filament material. The tips of the bristles may be flocked (split) or unflocked. The tips of the bristles may also be rounded, bulbous, flat, pointed, etc. The bristles may be soft and flexible for a comfortable and soothing treatment, or may be rigid and stiff for a more aggressive tissue treatment. In some embodiments, the bristles include both soft/flexible bristles and rigid/stiff bristles for a combined treatment. In other embodiments, a first brush of the treatment device **5500** may have a first type of bristles and a second brush having a different type of bristles.
(157) The therapy device of FIG. **55** may also include a light therapy system **5544** including a plurality of light emitting diode (LED) lights **5546** directed toward the skin to provide additional therapy to the fascia tissue. The light therapy system **5544** may provide light in one or more of the following forms: Red light (625 nm), Blue light (415 nm), Red+Blue light (625 nm-415 nm), Infrared (760 nm). The computing device may be configured to power and control operation of the LEDs. User input controls, such as knobs, buttons, or otherwise, may be accessible to a user from the treatment device **5500** and/or base station to control operation of the LEDs and control circuitry.
(158) The LEDs may be disposed at any location in which the light can be properly directed to a patient's skin (i.e., is not blocked by another element of the treatment device **5500**). For example, the LEDs may be at the tips of the tissue treatment element(s) of the tissue treatment assembly **5504** or along a periphery of the treatment device **5500**. The LEDs may be pointed at an angle outward from the device, toward the tissue treatment assembly **5504**, or in any other location, arrangement, and/or orientation so as to direct light onto the skin surface. The LEDs may be aimed in front of, to the side, or between the treatment elements so that the lights may be incident skin of a user prior to and/or after the treatment elements pass a region of skin of a user. In an embodiment, a motion sensor may sense when the device is in use (e.g., moving back and forth) and cause the LEDs to automatically turn ON during motion, and cause the LEDs to automatically turn OFF when not in motion or not moving in a particular treatment motion (e.g., rotating, substantially linearly forward/backward or side-to-side, where substantially means that there may also be some rotational and/or vibrational movement during operation). One or more pressure sensors (e.g., between the treatment element(s) and bar) may also be utilized to determine when the device is in operation and cause the LEDs to turn ON and OFF. A timing circuit may be utilized to maintain the LEDs in the ON state for a minimum duration of time (e.g., 15 seconds). In an embodiment, circuitry may turn the LEDs ON and OFF in a particular pattern, such as lighting certain LEDs when moving in a first direction and other LEDs when moving in a second direction.
(159) As shown in FIG. **55**, the treatment device **5500** may also include a soft tissue stimulation system **5548** configured to provide electrical current to the treatment area by placing a plurality of

electrodes on the skin surface and providing electrical impulses via the electrodes to the skin and soft tissue (such as fascia tissue) below the skin's surface. In some embodiments, the stimulation system **5548** employs circuitry and hardware elements that can execute traditional TENS (transcutaneous electrical nerve stimulation) and/or LAMES (neuromuscular electrical stimulation) therapy. In such embodiments, at least one lead wire may be electrically coupled to the treatment device **5500**, with a transcutaneous electrode at the distal end for delivering the electrical impulses to the patient. The transcutaneous electrode may adhere to the skin. The treatment device **5500** may be configured to provide a pre-determined stimulation waveform having a pre-determined frequency (Hz), pulse width (μs), and amplitude (mA). Alternatively the device may be configured to allow a user to modify one or more parameters of the stimulation waveform.
(160) In some embodiments, the electrodes may alternatively be positioned on the treatment device housing (such as on the frame, the bar member, the treatment elements, or the treatment accessory, to be placed in direct contact with the skin for stimulation. In an embodiment, the treatment accessories, for example, may be configured with an accessible compartment that is configured to store batteries, control circuitry, electrode(s), wires, etc., thereby enabling the treatment to be self-contained within the treatment accessories. In other embodiments, the treatment accessories are electrically coupled to the treatment device and powered by the base station. User input controls, such as knobs, buttons, or otherwise, may be accessible to a user to control operation of the stimulation signals applied to a user from the electrodes controlled by the control circuitry. In operation, the user may remove the electrode(s) from the compartment and apply to him or herself. The electrodes, in an embodiment, may be attached to straps that may wrap around or be applied to a patient's body, such as an arm or leg, so as to apply the TENS or LAMES treatment before, during, or after fascia tissue treatment by the treatment device **5500** with the treatment elements. As an example, the electrodes may be positioned to the sides of a pathway that the treatment elements are to be applied and electrical stimulation may occur before, during, or after treatment.
(161) As described above, the base station may be configured for use with multiple different types and/or styles of treatment devices (e.g., therapy devices, etc.). In some embodiments, the treatment plan may involve using different treatment devices and/or device attachments in a certain order. For example, the treatment plan may include application of a skin and/or soft tissue preparation device to the patient. The preparation device may include equipment that is structured to be worn by the patient. In one embodiment, the preparation device may include an outer shell (e.g., plastic, etc.) that is shaped to match the contour of a certain body area(s) (e.g., a patient's head, shoulder, thigh, etc.). An illustrative shell **5602** (e.g., armor, etc.) for a preparation device **5600** is shown in FIGS. **56-57**. An illustrative helmet **5803** (e.g., head gear, etc.) for a preparation device **5800** is shown in FIG. **58**. The preparation device **5600**, **5800** may include soft padding that is coupled to an interior surface of the shell **5602** and/or helmet **5803** to improve patient comfort, and straps (e.g., a hook and loop fastener, cord, etc.) to help secure the preparation device **5600**, **5800** to the patient during pre-treatment. The preparation device **5600**, **5800** may be configured to pre-heat and/or soften the patient's tissue prior to the use of other treatment modalities. The preparation device **5600**, **5800** may include lights (e.g., UV lights, infrared heating elements, etc.), gel-packs, ultrasonic transducer (e.g., vibrating elements), or another technology that is coupled to the shell **5602**, helmet **5803** and/or padding and positioned to engage with a patient's skin. In some embodiments, the preparation device **5600**, **5800** may also include nozzles and/or another dispensing device or system to apply lubricating oils or other agents to facilitate heat absorption to the skin and to prepare the skin/tissue for further treatment).
(162) After pre-treatment, in at least one embodiment, the treatment plan may involve using a radio frequency (RF) device (e.g., RF head) as an initial treatment operation. The RF device may be configured to use energy waves (e.g., thermal energy) to help stimulate the superficial skin layers and/or to heat the deep layer of a patient's skin (e.g., to within a range between approximately 122° F. and 167° F., or another suitable temperature). Illustrative RF treatment devices **5900**, **6000** are

shown in FIGS. **59-60**. The treatment plan may include using the RF treatment device(s) **5900**, **6000** to apply heat for a prescribed period to stimulate the tissue and promote creation of new collagen fibers.

(163) The treatment plan may also include using an ultrasound skin treatment to facilitate the removal of fat deposits under a patient's skin and/or heat the patient's tissue. For example, FIG. **61** shows an illustrative ultrasonic cavitation device **6100** for a tissue treatment that may be coupled to the tether and base station. The cavitation device **6100** may be configured to apply ultrasonic radio waves to the tissue, causing a disruptive vibration. Other example treatment device that may be used include vacuum devices that apply negative pressure to tissue during operation, and/or combination devices that utilize multiple treatment modalities (e.g., RF in combination with spinning or other movement). The treatment plan may also include application of infrared light (e.g., via an IR treatment device), pre-cancerous treatments (e.g., blue light therapy), acne treatment, and/or other tissue treatments. These treatment devices may be accessory heads that can be interchangeably coupled to a single treatment device or separate treatment devices that are coupled to the base station in place of the treatment device, as described above.

(164) The size of the treatment device and tissue treatment assemblies may also be different in various embodiments. For example, FIG. **62** shows another illustrative fascia tissue treatment device **6200** that is designed to treat larger body areas (e.g., the back, legs, stomach, etc.). FIGS. **63-75** show various illustrative fascia tissue therapy devices **6300**, **6400**, **6500**, **6700**, **6800**, **6900**, **7000**, **7100**, **7200**, **7300**, **7400**, **7500** that are designed to manipulate fascia tissue in smaller body areas or to focus application of treatment. These devices may have a smaller overall footprint that the treatment device **6200** of FIG. **62**. As shown in FIGS. **65**A-**65**D and FIG. **66**, the tissue treatment assembly **6504** (e.g., the effector, etc.) for these smaller devices may include a single tissue treatment element (e.g., a panel with fingers forming a single claw, etc.) to support the treatment element. As shown in FIGS. **74-75**, the size of the handles **7412**, **7512** and/or grip for the treatment device **7400**, **7500** may also differ depending on the size of the treatment device **7400**, **7500** and/or its intended use. In some embodiments, the treatment system (e.g., computing device) may be configured to receive data from remote systems, such as other tissue treatment systems, medical devices, and/or data repositories and may be configured to use the received data to determine and/or adjust a treatment plan for the patient.

(165) Referring to FIGS. **76-78**, an illustrative tissue treatment assembly **7600** (e.g., an effector, etc.) for a fascia tissue treatment device is shown. The tissue treatment assembly **7600** may be made from separate pieces that are fastened or otherwise coupled together. In other embodiments, the treatment assembly may be formed may be a monolithic body formed from a single piece of material (e.g., from plastic via an injection molding process, etc.). The treatment assembly **7600** may include a panel **7602** (e.g., a board, a plate, etc.) that supports a plurality of fascia tissue treatment elements that extend at least partly axially away from the panel **7602**. The panel **7602** may also include a support member **7604** (e.g., a support element, a quick-connect device, etc.) disposed on the panel **7602** that is configured to detachably couple the treatment assembly **7600** to an actuator of the treatment device.

(166) The support member **7604** is configured to engage the actuator to mount the treatment assembly **7600** to the actuator. The support member **7604** may include a connecting element **7605** (e.g., a shaft, a mount point, a coupler, etc.) that is configured to detachably couple the support member **7604** to the actuator, as will be further described. In some embodiments, the support member **7604** is coupled to the panel **7602** at a central position along the panel **7602**.

(167) In some embodiments, the support member **7604** may be configured to increase the stiffness of the treatment assembly **7600** and to reduce torque and stress on the actuator during operation. For example, as shown in FIG. **76**, the treatment assembly **7600** may include a plurality of support elements **7606** (e.g., support ribs, etc.) disposed between adjacent rings of finger members **7608** (e.g., rings of finger members **7608** extending in circumferential rows relative to a central axis

7610 of the panel **7602**) and/or aligned with the rings of finger members **7608**. The support elements **7606** may be disposed on a first surface (e.g., an upper surface, etc.) of the panel **7602**, opposite from the finger members **7608**. In some embodiments, support elements **7606** may also be included on a skin facing side of the panel (on the same side of the panel **7602** as the finger members **7608**, on a second/lower surface of the panel **7602**, etc.).

(168) The support member **7604** may be coupled to the panel **7602** at a position that is spaced radially apart from a center **7611** of the panel **7602**. For example, the support member **7604** may be arranged along the panel **7602** so that the connecting element **7605** is at least partially disposed at one of: (i) an outer perimeter **7612** of the panel **7602**, or (ii) an intermediate radial position **7614** between the center **7611** of the panel **7602** and the outer perimeter **7612** of the panel **7602**.

(169) As shown in FIG. **76**, the support elements **7606** include a plurality of cylindrical support ribs **7613** extending at least partially axially away from the panel **7602** and that are arranged concentric with a rotational axis (e.g., the central axis **7610**) of the panel **7602**. The cylindrical support ribs **7613** may be spaced radially apart from one another. In some embodiments, the support elements **7606** may also include planar ribs **7607** (e.g., spoke ribs, axial ribs, etc.) that engage with and extend between adjacent ones of the cylindrical support ribs **7613**. In some embodiments, the support member **7604** (e.g., the support elements **7606** together) form a skeletal framework **7616** that is engageable with a connector on the treatment device (e.g., the actuator, etc.) to ensure that the load is more uniformly distributed across the surface of the panel **7602**. The support elements **7606** (e.g., the cylindrical ribs) may include quick-connect fittings to attach the treatment assembly to the treatment device at the intermediate radial position **7614** along the panel **7602** or at along the outer perimeter **7612** of the panel **7602**, as described in further detail above.

(170) In the embodiment of FIGS. **76-78**, the panel **7602** is a disk-shaped plate that includes an opening **7618** (e.g., cut-outs, holes, etc.) at a location of each of the treatment elements (e.g., the finger members **7608**). The panel **7602** may have an outer diameter within a range between approximately 0.5 inches and 12 inches. In other embodiments, the shape and/or size of the panel **7602** may be different. The treatment elements are uniformly distributed across the panel **7602** at approximately equal intervals. As shown in FIGS. **76-78**, the treatment elements are arranged in concentric rows that are centered about a rotational axis (e.g., the central axis **7610**) of the panel **7602**. FIG. **79** shows another embodiment of a tissue treatment assembly **7900** that includes treatment elements (e.g., finger members **7908**) arranged in concentric rows relative to a central axis of a panel. In other embodiments, the treatment elements may be arranged in a spiral shape (e.g., as shown in the tissue treatment assembly **8000** of FIG. **80**) or another suitable shape. In some embodiments, adjacent rows of treatment elements are rotationally offset from one another (e.g., such that the treatment elements are not aligned along a radial direction, as shown in the tissue treatment assembly **8100** of FIG. **81**) to distribute stress more uniformly across the surface of the panel. However, it will be appreciated that the shape, size, and arrangement of treatment elements may be different in other embodiments.

(171) As shown in FIG. **77**, the treatment elements include a plurality of rigid finger members **7608** that are fixedly coupled to the panel **7602** at a proximal end **7620** of the respective finger members **7608** and extend away from the panel in at least partly axial direction. The plurality of finger members **7608** may be elongated so that a cross-sectional diameter of the finger members **7608** is less than a length of the finger members **7608**. The finger members **7608** may define a plurality of tip portions (e.g., tips, etc.) that are substantially co-planar to one another and are disposed along a reference plane that is parallel to and spaced apart from the panel **7602** and that is substantially perpendicular to the central axis **7610** of the panel **7602**. As shown in FIGS. **76-78**, the finger members **7608** may be at least partially curved and aligned or oriented along a direction of rotation of the panel **7602**. As shown in FIG. **77**, the finger members **7608** each include a first portion **7622** extending axially away from the panel **7602**, a second portion **7624** extending at an angle **7628** relative to the first portion **7622** (e.g., radially and/or circumferentially away from the first portion

7622, and a third portion **7626** extending partly axially away from the second portion **7624**. The third portion **7626** may be curved along a radial and/or circumferential direction (e.g., clockwise when viewed from above the panel **7602**) along the panel and/or about the central axis **7610** of the panel **7602**.
(172) In at least one embodiment, as shown in FIGS. **77-78**, each one of the finger members **7608** has a central axis **7630** (e.g., finger member axis, etc.) extending from the proximal end **7620** to a distal end **7621** of the respective finger member **7608**. At least one of the central axes **7630** may extend at least partly along a radial direction **7632** and/or a circumferential direction **7634** of the panel **7602** (e.g., relative to a cylindrical coordinate system **7631** that is centered on the proximal end **7620** of the respective finger member **7608** and/or relative to the central axis **7610** of the panel **7602**, etc.). The circumferential direction **7634** may be oriented along a direction of travel of the panel **7602** when the actuator is in the ON state so that at least a portion of each one of the finger members **7608** extends along the circumferential direction **7634**. In at least one embodiment, the central axes **7630** of the finger members **7608** extend along the circumferential direction **7634**. In some embodiments, the finger members **7608** extend away from the panel **7602** (e.g., are curved away from the panel **7602**) toward a direction of rotation of the panel **7602** when the actuator is in an ON state (e.g., so that the finger members **7608** extend along a first circumferential direction **7636** relative to the central axis **7610** that is the same as a direction of travel of the panel **7602** when the actuator is in the ON state). By aligning the finger members **7608** with the direction of travel, the finger members **7608** can project into a patient's tissue during treatment, in a scooping or shoveling motion to facilitate the release of fascia tissue. However, it should be appreciated that the actuator may be configured to rotate the panel **7602** in multiple directions (e.g., both the first circumferential direction **7636** or clockwise direction and a second circumferential direction **7637** or counterclockwise direction) in some embodiments, and that the opposing alignment of finger members **7608** with the direction of travel of the panel **7602** can provide different effects depending on the rotational direction, which can further promote fascia tissue release in some circumstances.
(173) The size, shape, and arrangement of finger members **7608** along the panel **7602** may be different in various embodiments. As shown in FIG. **76**, at least one of the finger members **7608** may include a shaft **7638** having a convex surface **7640** extending between the proximal end **7620** and the distal end **7621** of the at least one finger member **7608**. The at least one finger member **7608** may define a tip **7642**, which may be monolithic with the shaft **7638** and at which the convex surface **7640** transitions to a concave surface **7644**. The concave surface **7644** may be oriented with a direction of travel of the panel **7602** when the actuator is in the ON state so that the concave surface **7644** defines a leading surface/edge of the at least one finger member **7608**.
(174) In other embodiments, the size, shape, and arrangement of the finger members **7608** along the panel **7602** may be different. For example, the finger members **7608** may be rotated approximately 90 degrees from the orientation shown in FIGS. **76-78** such that the second portion **7624** and the central axis **7630** of the finger members **7608** extend at substantially along the circumferential direction **7634**.
(175) FIG. **82** shows an illustration of a side cross-sectional view of another illustrative tissue treatment assembly **8200** (e.g., an effector, etc.). As shown, the finger members **8208** each include a first portion **8222** engaged with a panel **8202** and of the treatment assembly **8200** extending axially away from the panel **8202**, and a second portion **8224** engaged with the first portion **8222** and curved along a circumferential direction, and along a rotational direction of the panel. In this arrangement, each of the individual finger members **8208** have a convex surface **8240** extending between a proximal end and a distal end of the respective finger member **8208** that faces axially toward the panel **8202**, so as to form small scooping elements having a smaller first radii **8246** along an inner surface **8248** than along an outer, skin facing surface **8250**. Beneficially, angling and/or curving the finger members **8208** along the circumferential direction can facilitate

manipulation of the fascia tissue layers and promote release and treatment of abnormal/damaged fascia tissue. Reversing the rotational direction of the panel **8202**, so that the distal end of the finger members **8208** define a trailing surface and/or trailing edge of the finger members **8208** may alter the forces applied to the fascia tissue during treatment and may further effectuate the release of fascia tissue in certain treatment modalities.

(176) As shown in FIG. **82**, in some embodiments, the panel **8202** of the tissue treatment assembly is substantially planar. In other embodiments, the panel **8202** may be made from a flexible material (e.g., a material capable of bending easily without breaking, a pliant material, a material able to bend to form a radius of curvature that is at least 5%, 10%, 15%, or greater of an outer radius of the panel **8202**, etc.) and may be curved along a radial direction in a neutral position of the panel **8202** (in which no forces are being applied to the panel **8202**) such that a central region or center **8212** of the panel **8202** (e.g., proximate to a rotational axis of the treatment assembly) is raised above or protrudes axially away from an outer perimeter **8213** of the panel **8202**. As the tissue treatment assembly **8200** is pressed toward (e.g., along an axial direction) a patient's skin during treatment, such as in response to an axial force applied to the center **8212** of the panel **8202**, the center **8212** of the panel **8202** bends toward the skin and into axial alignment with the rest of the panel **8202** (e.g., such that the center **8212** and the outer perimeter **8213** of the panel **8202** are disposed along the same reference plane and/or are substantially flush with one another). A thickness **8242** of the panel **8202** may be sized such that a force required to substantially the center **8212** with the outer perimeter **8213** of the panel **8202** is less than or equal to a target force applied to the skin during treatment (e.g., 1N, 5N, 10N, or any force between or greater than the foregoing forces).

(177) FIG. **83** shows a side cross-sectional view of yet another illustrative tissue treatment assembly **8300** (e.g., an effector, etc.). As shown, the treatment elements of the treatment assembly **8300** include a plurality of finger members **8308** that are curved back toward the panel such that at least one (or each) finger member **8308** engages the panel at opposing ends of the finger member **8308** (e.g., at both a proximal end **8320** and a distal end **8321** of the respective finger member **8308**). At least one finger member **8308** may include a cutoff (e.g., straight section) along an outer surface of the respective finger member **8308**, proximate to an end of the finger member **8308** such that the finger member **8308** includes a first edge **8340** extending substantially axially away from the panel **8302** and a second edge **8342** opposite from the first edge **8340** that curves back toward the panel **8302**. In such an arrangement, the finger member(s) **8308** “scoop” or otherwise act against the tissue along a rotational direction of the treatment assembly **8300** (e.g., along a single rotational direction when the first edge **8340** is a leading edge of the finger member **8308** during operation). As such, reversing the rotational direction of the treatment assembly **8300** may reduce how aggressively the finger members engage with the patient's tissue. In such an arrangement, concave surfaces of the finger members **8308** lead the finger members **8303** along the rotation of the finger members **8308**, thereby causing tips of the finger members **8308** (e.g., a point along the finger members **8308** that is farthest from the panel **8302**) to follow the shafts (e.g., main body portion, etc.) of the finger members **8303** rather than lead the shafts of the finger members **8308**. In some embodiments, by aligning the orientation of the finger members **8308** with the rotation direction, rotational stress of each of the finger members **8308** may be reduced.

(178) FIG. **84** shows a side cross-sectional view of yet another illustrative tissue treatment assembly **8400** (e.g., an effector, etc.). The treatment assembly **8400** includes a plurality of finger members **8408** that extend away from the panel at an oblique angle **8440**. As shown in FIG. **84**, the finger members **8408** are angled along a circumferential direction toward the direction of rotation of the panel when the actuator is in an ON state. As shown, the plurality of finger members **8408** may define respective tip portions **8442**, which may be substantially co-planar. The tip portions **8442** may define a plane **8444** (e.g., a tip-contact plane, etc.) that is spaced apart (e.g., axially, etc.) from the panel and that is substantially perpendicular to a central axis **8430** of the panel. In other embodiments, such as when the panel is made from a flexible material, the panel may bend under

an applied force to bring the tip portions **8442** into alignment with the plane **8444**. In yet other embodiments, such as during application to curved body areas, the panel may remain curved.
(179) FIG. **85** is a side cross-sectional view of yet another illustrative embodiment of a tissue treatment assembly **8500** (e.g., an effector, etc.) for a fascia tissue therapy system. As shown, the tissue treatment assembly **8500** includes a panel or base and multiple tissue treatment elements that are coupled to the panel. In at least one embodiment, the tissue treatment elements are finger members **8508** that extend axially away from the panel. In other embodiments, at least one finger member **8508** is angled with respect to a central axis of the panel. In yet other embodiments, at least one finger member **8508** curves away from the panel. The panel may include a circular, planar disc or another suitable support structure for the finger members **8508** (e.g., a skeletal framework, etc.). The finger members **8508** are shown as cylindrical elements but may have a different geometry or size in other embodiments.
(180) For example, the size of the finger members **8508** may increase with increasing radial position **8540** along the panel, relative to a central axis **8530** of the panel. In at least one embodiment, an outer diameter **8542** of the finger members **8508** increases (e.g., continuously) with increasing radial position **8540** along the panel. In this way, a contact area between each finger member **8508** and a person's skin increases with increasing radial position **8540** along the panel. This arrangement can reduce the risk of tissue damage in areas of the panel where the speed is greatest (e.g., approaching the outer diameter of the panel). By using different finger profiles based on distance from a center location, the treatment assembly may be used in different manners.
(181) FIG. **86** shows a side cross-sectional view of yet another illustrative treatment assembly **8600** (e.g., an effector, etc.). The treatment assembly **8600** includes finger members **8608** that have different shapes. The shape of the finger members **8608** may vary with radial position **8640** along the panel, relative to a central axis **8630** of the panel. As shown in FIG. **86**, the finger members **8608** are arranged in concentric rows that each extend along a circumferential direction, and the shape of the finger members **8608** varies between the rows. In other embodiments, the shape of the finger members **8608** may also vary within a single row (e.g., along a circumferential direction within the row, etc.). The finger members **8608** may taper to form different sized and/or shaped finger member tips **8642** at a distal end (e.g., outermost end, etc.) of the finger members **8608**. A size of the finger member tips **8642** may increase with increasing radial position **8640** along the panel such that outer finger members **8608** can perform slightly different treatment than the inner finger members **8608**. In the embodiment of FIG. **86**, an innermost set of finger members **8608** (e.g., tapered finger members, etc.) nearest a central axis **8630** of the panel have a substantially triangular shape (with a rounded or curved tip), whereas an outermost set of finger members **8608** (e.g., untampered finger members, etc.) farthest from the central axis **8630** of the panel have a substantially rectangular shape (e.g., a rectangular body with a curved lower surface or face that forms a tip at a tip region).
(182) It should be appreciated that the shape of the finger members **8608** may be different in other embodiments. For example, the finger members **8608** may be cylinders, polygons, or have any other suitable cross-sectional geometry. The finger members **8608** may include at least one finger member **8206** that rotates or otherwise moves independently from the panel. For example, as shown in FIG. **92**A, a tissue treatment assembly **9200** may include at least one finger member **9202** that includes, or is formed as, a roller ball (e.g., a spherical element, etc.) that may rotate independently from the panel (e.g., in a clockwise direction as shown in FIG. **92**A, in a counterclockwise direction, etc.). In other embodiments, the roller ball(s) may be stationary with respect to the panel. The roller ball(s) may be included with other finger members of various alternative geometries, thereby enabling the other finger members to perform the fascia tissue treatment and the roller ball(s) to support movement of the finger members **8608** during treatment, for example. In other embodiments, as shown in FIG. **92**B, a tissue treatment assembly **9204** may include a plurality of roller ball elements spaced apart along the panel. Referring again to FIG. **86**,

the finger members **8608** may be tapered and have a larger, first cross-sectional area **8641** near the panel (e.g., at a proximal end of the finger member **8608**) and a smaller, second cross-sectional area **8643** away from the panel (e.g., at a distal end of the finger member **8608**). For example, as shown in FIG. **92**C, at least one finger member (e.g., treatment element **9208**) may be shaped as a nugget tip or may be a straight cylindrical finger nub as shown in FIG. **85**. In some embodiments, as shown in FIG. **87**, the finger members **8708** are individual elements that are spaced apart across the face of the panel.

(183) The finger members may be formed separately from the panel as individual elements that are fastened or otherwise coupled to the panel, or integrally formed with the panel from a single piece of material (i.e., monolithic with the panels). In other embodiments, at least two of the finger members may be integrally formed with one another from a single piece of material. For example, at least two finger members may be formed in a sinusoidal shape from a single piece of material via an injection molding process or another suitable manufacturing process. In some embodiments, the treatment assembly may include multiple subpanels coupled thereto. Each subpanel may include a plurality of finger elements. The subpanels may be circular panels having a smaller diameter than the panel, or may be rings that form a single continuous row of finger members of the panel. The subpanels may be removably coupled to the panel and arrangements of subpanels may be tailored to the treatment needs of the patient.

(184) In some embodiments, the treatment assembly includes other elements beyond the finger members to facilitate or enhance the effectiveness of the treatment. For example, FIG. **87** shows a side cross-sectional view of yet another illustrative treatment assembly **8700** that includes a plurality of finger members **8708** (e.g., a first plurality of treatment elements) and a plurality of secondary treatment elements **8740** in between the finger members **8708**. In at least one embodiments, the finger members **8708** and secondary treatment elements **8740** are arranged in alternating concentric rows along the panel (e.g., where each “row” extends along a circumferential direction relative to the central axis of the panel). In other embodiments, the finger members **8708** and secondary treatment elements **8740** are positioned along the same row (e.g., in alternating arrangement along the row, etc.). It should be appreciated that any other arrangement of finger members **8708** and secondary treatment elements **8740** may be used to improve treatment performance.

(185) The secondary treatment elements **8740** may be any form of skin treatment element or device. For example, the secondary treatment elements **8740** may include brush elements **8742** that are configured to engage the skin to remove particles (e.g., dirt, oil, dead tissue, etc.) and/or to facilitate application of a lubricant, lotion, and/or other tissue treatment solution. The brush elements **8742** may also perform other functions such as relaxing a patient's skin at a treatment site, performing micro-dermabrasion, or another use. The bristles of the brush elements **8742** may be made of any suitable material including nylon, polypropylene, horse hair, feather, or any other natural or synthetic filament material. The tips of the bristles for at least one brush element **8742** may be flocked (split) or unflocked. The tips of the bristles may also be rounded, bulbous, flat, pointed, etc. The bristles may be soft and flexible for a comfortable and soothing treatment, or may be rigid and stiff for a more aggressive tissue treatment. In some embodiments, the bristles include both soft/flexible bristles and rigid/stiff bristles for a combined treatment. In other embodiments, a first brush of the treatment device may have a first type of bristles and a second brush having a different type of bristles. The brush elements **8742** may also have different types of bristles in the same brush elements **8742**.

(186) In some embodiments, the secondary treatment elements **8740** form part of a soft tissue stimulation system configured to provide electrical current to the treatment area by placing (i) a plurality of electrodes on the skin surface and (ii) providing electrical impulses via the electrodes to the skin and soft tissue (such as fascia tissue) below the skin's surface, as described above. In some embodiments, the stimulation system employs circuitry and hardware elements that can execute

traditional TENS (transcutaneous electrical nerve stimulation) and/or NMES (neuromuscular electrical stimulation) therapy. In such embodiments, at least one lead wire may be electrically coupled to the device, with a transcutaneous electrode at the distal end for delivering the electrical impulses to the patient. The transcutaneous electrode may adhere to the skin. The device may be configured to provide a pre-determined stimulation waveform having a pre-determined frequency (Hz), pulse width (μs), and amplitude (mA). Alternatively the device may be configured to allow a user to modify one or more parameters of the stimulation waveform.

(187) In some embodiments, the electrodes may alternatively be positioned on the panel (e.g., in between finger members along each row, in separate rows from the finger members, etc.). In other embodiments, the electrodes may be positioned on the finger members themselves (e.g., at least one finger member may include an electrode positioned thereon).

(188) Referring to FIGS. **88-91**, various additional illustrative treatment assembly (e.g., effector, etc.) designs are shown. In the embodiment of FIG. **88**, the treatment assembly **8804** includes a single tissue treatment element (e.g., flower member, claw, etc.) having a central body **8807** and a plurality of finger members **8808** extending axially and curving radially away from the central body **8807**. FIGS. **89-91** each show an illustrative treatment assembly **8900**, **9000**, **9100** that includes a panel and a plurality of tissue treatment elements coupled to the panel and extending axially away from one side of the panel. The tissue treatment elements may be arranged in at least one ring concentric with a rotational axis of the effector, arranged in rows along a radial direction, or any other suitable arrangement. In one embodiment, and as shown in FIG. **89**, the treatment elements are mounted to the panel using a fastener **8940** (e.g., screw, bolt, etc.) extending through the central body of each individual tissue treatment element. In other embodiments, the tissue treatment elements may be glued, welded to the panel, or otherwise coupled to the panel. In yet other embodiments, the tissue treatment elements may be integrally formed with the panel from a single piece of material. The treatment elements, including the finger members, may be formed of metal or another material that may be sanitized after use without damage. In other embodiments, the treatment elements may be formed from acrylic, PVC, hard rubber, or any other material that is stiff and does not cut or scrape skin of a patient on which the effector is being utilized to help treat or adjust fascia tissue. It will be appreciated that alternative numbers of treatment members may be utilized in accordance with the principles of the present disclosure. The treatment elements are also shown to be substantially identical. However, it will be appreciated that alternative configurations of the treatment elements may be utilized to provide for treating different size anatomical regions.

(189) The treatment elements shown in FIG. **89** have an outer diameter **8942** of approximately 1½ inches. However, the outer diameter **8942** of the treatment elements may have a fairly wide range (e.g., ½ inch to 6 inches in diameter, etc.). Illustrative treatment elements shown have a length **8944** about ¾ of an inch long and have heads or tips that have a head diameter **8946** of about ⅜ of an inch. The dimensions and configurations (e.g., curves) of the treatment elements, finger members, and tips of the finger members may vary depending on the anatomical region on which the treatment assembly is to be used. The tips of the finger members may have one or more same or different dimensions as the finger members (e.g., the tips may have a larger diameter by being bulbous). The finger members are shown to be curved. Alternative configurations, such as finger members being straight may be utilized as well. The treatment elements are each shown to be a single member. However, in other embodiments, the treatment elements may be formed from multiple elements. Still yet, rather than the treatment assembly using treatment elements that have a flower, claw, or cactus-like appearance (i.e., central portion with extending finger members), treatment elements with non-flower-like appearance may be utilized, as well, that still provides a user with a number of closely spaced pressure-point elements that can be pressed and guided along a patient's skin to cause fascia tissue to be released or perform a non-therapeutic function. The finger members may be substantially the same length (e.g., less than 0.1 inch difference in length between finger length) such that the tips of the finger members are substantially co-planer and

parallel to a support structure (in the case of a flat support structure) so that a pressure load applied to the skin and fascia tissue is substantially equally applied by each of the finger members.

(190) Each of the treatment elements is shown to have six finger members. Alternative numbers of finger members may be utilized in accordance with the principles of the present disclosure. The finger members may be stiff or rigid (e.g., inflexible, etc.), thereby having minimum bend or deformation during usage of the device on fascia tissue of a patient.

(191) The panel may have openings (not shown) defined by the panel through which a screw or other fastening mechanism may extend through treatment elements into the panel. After fastening the treatment elements to the panel, glue or other fastening material, such as epoxy, may be utilized to secure the treatment elements to the panel. A cover (not shown) above the fastening mechanisms may be utilized to limit the ability for someone to access or remove the fastening mechanisms of the treatment elements. Alternatively, the treatment elements may be configured to allow for a user to more easily replace the treatment elements to change size, replace broken flower members, or otherwise.

(192) In an embodiment, the treatment assemblies **8900**, **9000**, **9100** may be configured such that the tissue treatment elements themselves may spin by themselves or in addition to the panel spinning or being stationary. In an embodiment, the tissue treatment elements may be mounted on rotatable shafts that extend through the panel and are mounted to hubs that secure the shafts to the panel such that the tissue treatment elements are maintained at a fixed vertical position.

(193) Gears may be mounted to the rotatable shafts to cause the tissue treatment elements to spin as other gears apply rotational force to the gears connect to the rotatable shafts. The other gears may be driven by separate motors, a common motor, the same motor that drives the panel to rotate, or otherwise. In an embodiment, all of the tissue treatment elements may be configured to rotate at the same speed. In an alternative embodiment, a portion of the tissue treatment elements may be configured to rotate. In yet another embodiment, some or all of the tissue treatment element(s) may be configured to rotate at different speeds. In an embodiment, rotation of the tissue treatment elements may be selectively turned ON or OFF. In other words, an operator may be able to engage or disengage rotation of tissue treatment elements, as desired. In a stationary or OFF state, the panel may spin or rotate and the tissue treatment elements may remain in a fixed position relative to the panel. In an ON state, the tissue treatment elements may rotate relative to the panel. In an embodiment, speed of rotation may be constant or a motor that drives the gears to which the tissue treatment elements may be selectably altered to operate at different speeds, which may or may not be related to the speed of rotation of the panel. Gear ratio for driving the treatment elements may be set to a speed ranging from 1 revolution per minute (RPM) to 1000 RPMs. Alternative rotational speeds of the tissue treatment elements may be utilized. The gears may be singular or allow for multiple gear ratios.

(194) The tissue treatment elements may be in the shape of a claw having a base with finger members extending therefrom. It should be understood that alternative configurations of the finger members may be utilized. The shapes may have a wide range of geometries that may be configured to perform “fascial shearing” for different layers of fascia tissue. In an embodiment, a tight latex or other material wrap may be applied to the body part in performing the process. The different layers of fascia tissue may be based on the anatomical region at which the fascia tissue is located. For example, fascia tissue on a person's scalp is much closer to the person's skin than fascia tissue on the person's leg or torso. Moreover, the physical dimensions of the patient may also necessitate tissue treatment elements of different configurations. For example, if the patient has minimal body fat (e.g., less than 10% body fat), then the tissue treatment elements may be shorter than if the patient has a lot of body fat (e.g., over 50% body fat). Moreover, a patient with higher density of muscle (e.g., young male who in good physical condition) may also demand tissue treatment elements that have tips with a broader circumference than a patient with less density of muscle (e.g., senior female). Hence, the configuration of the tissue treatment elements may vary

considerably for different patients.
(195) Although tissue treatment elements that are rigid with rigid finger members (i.e., finger members and base members) may be utilized in some embodiments, finger members with some amounts of compliance (i.e., bendable or rotatable) are possible. Such compliant finger members may be possible with finger members formed of certain flexible material (e.g., rubber, silicone, etc.) or are mounted to the base member via a flexible member of attachment mechanism. Although some compliance may be provided, the tissue treatment elements may be configured to treat fascia tissue at particular anatomical regions and possibly of particular type. For example, fascia tissue on a person's forehead may be treated with tissue treatment elements with different configurations than fascia tissue on a person's cheek or neck. Although the finger members shown have an arcuate shape extending between the proximal end at the panel and distal end at the tip with a fairly uniform diameter shaft between the proximal and distal ends, it should be understood that a wide variety of shapes of finger members may be utilized. For example, the diameter between the proximal end and tip may vary from a first diameter at the proximal end to a second diameter at the distal end, where the first diameter is larger than the second diameter. The transition between the first and second diameters may be smooth with no discontinuities or the transition may be more abrupt (e.g., step between two or more diameters along the length of the shaft of the finger members).
(196) Curvature or other configuration of the finger members may vary between different treatment assemblies such that an operator may select from amongst several different treatment assemblies based on the patient (e.g., high body fat versus low body fat), depth of fascia and/or type of fascia at an anatomical region being treated, speed of the rotation of the treatment assembly. It should be understood that the various configurations of the finger members being described with regard to the tissue treatment elements may be applied to finger members that are directed connected to a panel, such as shown in FIGS. **76-87**.
(197) Referring to FIG. **92**, another illustrative treatment element structure is shown (treatment element **9208**) that may be used in place of one or more finger members of the treatment assembly, and/or as a standalone treatment element **9208** that may be used by the treatment device to manipulate fascia tissue. In the embodiment shown, the treatment element **9208** is a refining tool that is used to target small or stubborn areas of fascia tissue. The treatment element **9208** includes a base **9240** and a conically-shaped extension **9242** engaged with and extending away from the base **9240**. The treatment device (e.g., actuator) may be configured to press a tip of the conically-shaped element into a patient's tissue and then move the treatment element **9208** along a circular path to work the affected area. In other embodiments, the treatment device (e.g., actuator) may be configured to rotate the tip of the treatment element **9208** by pivoting the element from the base **9240**. It will be appreciated that various treatment element shapes and sizes may be used in other embodiments.
(198) While certain features of the treatment device (e.g., therapy device, etc.) are configured to be optimal usage on fascia tissue, the features also provide for ornamental appearance. It should be understood that the treatment device may be used for increasing overall myo-fascial fitness to loosen fascia tissue that is constrained, improve health and/or beauty purposes (e.g., provide a satisfactory feeling to a user and/or alter the appearance of cellulite and skin smoothness). Moreover, usage of the fascia tissue fitness device may open, loosen, restore, and/or revitalize fascia tissue of men and women, young and old. FIGS. **96-100** show photographs demonstrating some of the various benefits that can be provided by using the treatment system (e.g., therapy system, etc.). For example, FIG. **93** shows a significant reduction in the amount of cellulite **9300** presenting through a patient's skin along a leg after treatment. The treatment system can also be used to treat acne **9400** (FIG. **94**), scar tissue **9500** (FIG. **95**), and/or other dermatological conditions. FIG. **96** shows improvements in patients that are suffering from scoliosis **9600** and/or other forms of postural misalignment. The treatment system can be used to redistribute fatty tissues

9700 (FIG. **97**), promote and restore hair growth (FIG. **98**) (e.g., improving blood circulation to hair follicles **9800**, reducing stress on tissue from realignment of the fascia layers, etc.), restructure and remodel fascia (e.g., neck skin **9900**, etc.) to improve skin tightening (FIG. **99**) in different body areas, and to stimulate neurological connectivity and/or improve motor control for patient's suffering from paralysis **10000** (e.g., as a result of a stroke as shown in FIG. **100** or other injury). For people with underlying health conditions, such as diabetes, the system may be used to improve circulation to extremities, such as feet and hands, to improve circulation thereto. As shown in FIG. **119**, before and after photos of (i) a foot **19000** having poor circulation such that necrosis **19002** has set in on the big toe **19004** and skin **19006** on the remainder of the foot is showing significant cellular damage, and (ii) the foot **19000** after fascia tissue treatment such that circulation has improved and the skin on the big toe **19004** and rest of the foot **19000** shows significant amounts of improvement to the point that amputation has been averted.

(199) While various embodiments of the treatment system have been described in the context of use in medical environments, it will be appreciated that the treatment system may be also used in other applications. For example, the treatment system could be installed in a workout and/or training facility and used as part of a person's workout regimen, to improve physical appearance, stimulate blood flow, and/or improve recovery times. A miniaturized version of the treatment system could also be used for home use or in various other applications. Data from the computing device could be transmitted wirelessly to a personal device (e.g., smart phone, tablet, etc.) for additional review and analysis through applications installed on the user device (e.g., software as a service, etc.). The application may be configured to ensure compliance with health information privacy policies (e.g., the health insurance portability and accountability act (HIPPA), etc.) and may limit user access to certain data.

(200) FIGS. **101**A-**101**F are illustrations of perspective, another perspective, top, side, front, and bottom views of yet another illustrative fascia tissue treatment system **10100**.

(201) FIGS. **102**A-**102**G are illustrations of perspective, another perspective, top, side, front, and bottom views of yet another illustrative fascia tissue treatment system **10200**.

(202) FIGS. **103**A-**103**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue treatment device **10300**.

(203) FIGS. **104**A-**104**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative tissue treatment assembly **10400**.

(204) FIGS. **105**A-**105**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue treatment device **10500**. The fascia tissue treatment device **10500** may include LED lights that are used for heating up fascia tissue of a patient.

(205) FIGS. **106**A-**106**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustration fascia tissue treatment device **10600**.

(206) FIGS. **107**A-**107**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue treatment device **10700**. In this case, the fascia tissue treatment device **10700** may be an ultrasound device configured to generate (and optionally receive) ultrasound signals. The ultrasound signals may be used for causing fascia tissue to be stimulated, thereby loosening up the fascia tissue prior to or during treatment. If the ultrasound device includes an ultrasound sensor, then ultrasound images of fascia and other tissue may be captured by the sensor and used to manage current and future treatments, for example, along with storing images for a medical professional to review current status, progress from previous treatments, and projecting and planning future treatments.

(207) FIGS. **108**A-**108**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue treatment device **10800**. In this case, the tissue treatment element of the treatment device **10800** includes a single “claw” that includes fingers that have an arc so as to apply vertical pressure to skin of a patient, thereby allowing for deeper penetration into fascia tissue. Alternative sizes and numbers of claws may be utilized. In an

alternative embodiment, rather than the fingers extending from side walls of a central extension member, fingers may extend directly from a base plate and maintain a same curved, vertical orientation for applying pressure to a patient.
(208) FIGS. **109**A-**109**G are illustrations of bottom, front, perspective, left, rear, right, and top views of yet another illustrative fascia tissue treatment device **10900**. In this case, the fascia tissue treatment device **10900** may be an RF device that include RF LEDs that heat up fascia tissue of a patient. The treatment device **10900** may also include an ultrasound signal generator, thereby enabling a combination of treatments (i.e., RF and ultrasound). It should be understood that additional treatment signaling, such as low current signals, may also be applied by the fascia tissue fitness device, and an operator may select one or more of the treatments to be performed simultaneously.
(209) FIG. **110** shows a perspective view of yet another illustrative fascia tissue treatment device **11000**. The treatment device **11000** is configured to rotate a panel **11020** (e.g., a tubular member, etc.) on which finger members **11008** are connected that extends at least partially along an axial direction so as to move the finger members **11008** toward and away from the surface of the patient's skin so that only a portion of the finger members **11008** may be engaged with the skin at an instant in time.
(210) As shown in FIG. **110**, the treatment device **11000** may include a base **11011** including handles **11012** and a movable and rotatable element **11013** supported by the base **11011**. In one embodiment, the movable element **11013** is disposed between the handles **11012**. In other embodiments, the base **11011** includes only a single handle **11012** on one side of the movable element **11013**. The handles **11012** may be tubular or have any other configurations to enable a user (e.g., treatment professional) to handle and maneuver the device while performing a treatment. The shape of the treatment device **11000** may allow for treatment in areas that are more difficult to access using other treatment assembly (e.g., effector, etc.) designs.
(211) The movable element **11013** may include a cylindrical sleeve that extends axially in between the handles **11012**. The sleeve is rotatably coupled to or supported by the base **11011** and is arranged coaxially with the base **11011** and handles **11012**. The movable element **11013** also includes a plurality of finger members **11008** that are coupled to the sleeve and extend away from an outer surface of the sleeve. The finger members **11008** may extend radially or at an angle, and may be straight or curved (e.g., the finger members **11008** may extend along a circumferential direction relative to a central axis of the panel, etc.). In at least one embodiment, the finger members **11008** are arranged in substantially linear rows along the sleeve (e.g., between opposing ends of the sleeve). In other embodiments, the finger members **11008** may be staggered or disposed in any other desired arrangement along the sleeve. In the embodiment of FIG. **101**, the finger members **11008** curve away from a direction of rotation of the sleeve, along a circumferential direction (e.g., counterclockwise when viewed from the side of the sleeve). In other embodiments, the finger members **11008** may curve toward the direction of rotation (e.g., counterclockwise when viewed from the side of the sleeve). It should be appreciated that the direction of rotation of the sleeve may be controlled by a user (e.g., clockwise or counterclockwise depending on the desired treatment and/or user preferences).
(212) The finger members **11008** may be formed separately from the sleeve and coupled to the sleeve using screws, adhesive products, and/or another suitable fastener. In other embodiments, the finger members **11008** may be integrally formed with an outer portion of the sleeve from a single piece of material. For example, the finger members **11008** may be injection molded or otherwise formed onto a flat surface of a formable material, such as plastic, that can be heated and rolled back onto itself to form the outer portion.
(213) As the sleeve rotates, each row of finger members **11008** moves in a circumferential direction around a central axis of the sleeve to periodically engage the patient's skin. The cylindrical roller arrangement of the movable element **11013** allows for approximately uniform application of

treatment across large body areas such as the legs and back. The cylindrical roller arrangement also contours to body areas near joints, such as the back of the leg at the knee and/or the elbow pit on an anterior side of a patient's arm.

(214) As shown in FIG. **110**, the base **11011** may include a user interface to facilitate control of the fascia tissue treatment device **11000**. The user interface may include actuator(s) (e.g., switches, knobs, capacitive sensors, levers, etc.) to allow the user to control rotation of the sleeve relative to the base **11011** (e.g., operating speed, torque, rotational direction, etc.).

(215) FIG. **111** shows a side cross-sectional view through the sleeve and the base **11011**. The sleeve defines a cavity **11040** that may be tubular shaped and that extends longitudinally between the handles **11012**. A central portion of the base **11011** is disposed within the cavity **11040**. The sleeve completely surrounds (e.g., circumscribes, etc.) the base **11011** along a central portion of the base **11011** between the handles **11012**. In some embodiments, the treatment device **11000** includes at least one bearing (e.g., ball bearing, roller bearing, etc.) engaged with the base **11011** and the sleeve to support the sleeve and allow movement of the sleeve relative to the base **11011**. The bearings may be sized to accommodate radial loading due to applied pressure between the sleeve (e.g., finger members) and a patient's skin. In some embodiments, the bearings may be sealed and/or self-enclosed bearings to eliminate the need for periodic lubrication and to prevent ingestion of dirt or other debris.

(216) Continuing with FIG. **111**, the treatment device **11000** may further include an actuator **11006** to cause rotation of the sleeve with respect to the base **11011**. The actuator **11006** may be an electric motor that is separate from the base **11011** and the sleeve. The electric motor may be a direct drive motor **11042** (e.g., a brushless direct current (BLDC) motor, an AC synchronous motor if powered via tether to the base station, or another suitable motor type) that engages the sleeve and/or base **11011** without any intervening transmission or gear set. In other embodiments, the actuator **11006** includes another form of electromechanical, electromagnetic, pneumatic, hydraulic, or other device. In at least one embodiment, the actuator **11006** is disposed at least partially within the cavity **11040** or at the end(s) of the sleeve and is at least partially concealed by the sleeve. In other embodiments, the actuator (e.g., the electric motor) is at least partially formed by the sleeve and the base **11011**. For example, the sleeve may form part of a rotor **11044** of the actuator **11006** and the central portion of the base **11011** at least partially forms a stator **11046** of the actuator **11006**.

(217) In some embodiments, the treatment device **11000** further includes a control system to selectively power electrical coils of the stator **11046** and/or rotor **11044** based on the relative position between the sleeve and the central portion. The control system may include a controller and at least one sensor (e.g., a Hall Effect sensor, optical sensor, etc.) coupled to the controller to determine a rotational position of the sleeve relative to the stator **11046**. The control system may further include a user interface to control operation of the treatment device **11000**. As shown in FIG. **110**, the user interface may be disposed on at least one handle **11012** of the treatment device **11000**. In other embodiments, the user interface may be integrated with or accessible from the control system of the base station.

(218) In embodiments where the treatment device **11000** is portable and/or detachable from the base station, the treatment device **11000** may further include a power source (e.g., battery), which may be disposed within the handles **11012** or in the cavity **11040**. Beneficially, integrating the motor with the sleeve and the base **11011** can reduce the size and weight of the powered treatment device **11000**. The direct drive motor configuration can also reduce heat produced by the treatment device **11000** and wear on the internal components. In other embodiments, power via a cable may be connected to one or both handles **11012** of the treatment device **11000**.

(219) It should be appreciated that the geometry of the device (e.g., sleeve, finger members, etc.) may be different in other embodiments. For example, FIG. **112** shows yet another illustrative powered tissue treatment device **11200** that includes finger members in the form of tapered nubs

that extend substantially radially away from the sleeve (instead of curving along a circumferential direction as described with reference to FIG. **110**). In other embodiments, the shape, size, and/or position of the finger members along the sleeve may be different. In yet other embodiments, the sleeve may include a flexible belt that is fastened around the base or an outer roller for the sleeve. In this way, other shapes can also be used for the sleeve (e.g., an oval shape, racetrack shape, etc.).
(220) FIGS. **113-116** show various illustrative powered tissue treatment devices that are configured to facilitate treatment of other (e.g., non-planar, etc.) body areas that might be difficult to access with a cylindrical or planar interface. For example, FIG. **113** shows an illustrative powered treatment device **11300** that is configured to facilitate treatment of curved body areas (e.g., near the arm pit, neck, shoulders, joints, etc.). The treatment device **11300** may be the same as or similar to the illustrative fascia tissue treatment devices described with reference to FIGS. **59-75**, but include a different treatment assembly (e.g., effector, etc.) design.
(221) As shown in FIG. **113**, a treatment assembly **11304** of the treatment device **11300** includes a panel **11302** (e.g., a main body, etc.) and a plurality of finger members engaged with and extending away from the panel **11302**. In at least one embodiment, the panel **11302** is shaped as half of an ellipsoid (e.g., an elongated or stretched sphere) having a curved distal end (e.g., outer end) and that tapers between the proximal and distal ends (e.g., so that an outer diameter of the main body reduces continuously between the proximal and distal ends). In some embodiments, a central axis of the panel **11302** is co-linear with a rotational axis of the treatment device **11300** (e.g., a powered treatment head portion of the treatment device **11300**).
(222) The finger members may be coupled to the panel **11302** and extend away from the panel **11302**. The finger members may be curved along (or opposite to) a rotational direction of the treatment assembly **11304**. The finger members may be arranged in rows that extend along a circumferential direction around the panel **11302** or positioned in any other suitable arrangement. The shape and/or size of the finger members may also differ in other embodiments. For example, the FIG. **114** shows an embodiment of a tissue treatment assembly **11404** that is shaped as a geodesic polyhedron that includes triangular protrusions **11440** and indentations **11442** along an entire outer surface of the panel **11402**. The protrusions **11440** may be continuous along the outer surface, spaced apart at periodic intervals, or positioned in any other suitable arrangement.
(223) In other embodiments, the finger members may include triangular protrusions **11440** (e.g., tips), rounded protrusions, dimpling, and/or another suitable shape. For example, FIG. **115** shows a tissue treatment assembly **11504** that includes rounded spherical protrusions **11508** that extend away from an outer surface of the panel **11502**.
(224) The size and/or shape of the panel may also be different in other embodiments. For example, FIG. **116** shows an illustrative powered tissue treatment device **11600** that includes an elongated tissue treatment assembly **11604** (e.g., an elongated effector, etc.) having a longitudinal length that is approximately equal to a longitudinal length of the treatment head (e.g., housing, etc.). The increased length of the panel can facilitate manipulation of larger curved areas such as behind the knee or within an arm pit. The treatment assembly may also include finger members that are elongated in an axial direction and that form a substantially sinusoidal pattern across the outer surface of the panel (and extending along the axial direction). In other embodiments, the pattern formed by the finger member may be different (e.g., wave crests, sawtooth with rounded tips, etc.).
(225) FIG. **117** shows yet another illustrative embodiment of a powered fascia tissue treatment device **11700**. The treatment device **11700** includes a powered treatment head, which may be the same as or similar to any of the powered treatment heads/devices described herein. The treatment assembly **11704** includes a framework **11702** instead of a panel to support the tissue treatment elements. The framework **11702** has a circular outer shape but may be shaped differently in other embodiments. The framework **11702** may be lighter than a panel, thereby reducing load on a motor of the device used to rotate the framework **11702**. The framework **11702** includes a plurality of concentric ring elements **11740** (e.g., ring plates, etc.) that are arranged concentrically about a

rotational axis of the powered treatment assembly **11704** (e.g., a central axis of the framework). The framework **11702** may also include a plurality of support elements **11742** (e.g., ribs, radial extensions, etc.) that extend radially (or otherwise) between adjacent ones of the ring elements **11740** and support the ring elements **11740**. Although the powered treatment head is shown as being coupled to a central support element (e.g., central panel) of the framework **11702**, it should be appreciated that the powered treatment head may be coupled to other parts of the treatment assembly **11704**. For example, the powered treatment head may be directly coupled to a concentric ring element **11740** (e.g., the inner most ring, the outer most ring, etc.). Connecting the treatment head directly to the treatment assembly **11704** at a greater radial position along the treatment assembly **11704** can improve structural support and provide torque farther out from the center of the treatment assembly **11704** (thereby reducing stress on the support elements of the treatment assembly **11704**). As shown in FIG. **118**, the treatment elements may include a plurality of finger members that are engaged with and extend away from the framework **11702**. The finger members may be engaged with the concentric ring elements **11740** along both an inner and outer perimeter surface of the ring elements **11740**. The finger members may be arranged in opposed pairs that are radially aligned on either side of the ring elements **11740**. The finger members may extend at least partially radially away from the ring elements **11740** and also axially away from the framework **11702** (e.g., substantially parallel to a central axis of the framework **11702**). The finger members protrude radially beyond and inwardly from the ring elements **11740**. In some embodiments, the ring elements **11740** may be removably coupled (e.g., via screws, bolts, or another suitable fastener) to the support elements **11742** so that they can be individually removed and replaced. In other embodiments, at least one of the ring elements **11740**, support elements **11742**, and treatment elements are integrally formed from a single piece of material.

(226) FIG. **120** is an illustration of an inter-vaginal fascia tissue treatment device **12000** including various illustrative treatment effectors **12002**, **12004**, and **12006**. The device **12000** may be configured with a motor **12008** configured to rotate a shaft **12010** on which each of the treatment effectors **12002**, **12004**, and **12006** may be detachably connected for performing inter-vaginal fascia tissue treatment. The motor **12008** may be configured to rotate the shaft **12010** between 1 RPM and 1000 RPM, and a motor controller **12011** may be configured to control the spin of the motor **12008** in a variety of spin profiles (e.g., spline start and finish, linear increase/decrease, parabolic increase/decrease, etc.), such that the patient is safe during treatment. The treatment may include treating fascia tissue that is located along and adjacent to the vaginal walls of a patient. For example, it is possible for women to suffer certain types of injuries or have natural occurrences that distort and/or cause adhesions in the fascia tissue within the vagina region. In other words, fascia tissue in the vagina walls or region accessible via the vagina may become unstructured, thereby causing pain or discomfort during sex or other physical activities. Once such type of injury may occur during child birth as vaginal tears or perineal lacerations may occur. In the event of a tear to the vaginal wall, injury to the transverse perineal muscles (e.g., tear to fascia sheath or muscle), and/or fascia tissue thereat, adhesions or other misalignment of the fascia tissue may occur and scar tissue may be created. For any of these abnormalities, the tissue treatment device **12000** along with the various effectors **12002**, **12004**, and **12006** may be utilized for treatment.

(227) The tissue treatment device **12000** may be handheld and be powered with a power cord **12012** to charge a battery **12014** that may be used to power the motor **12008**. In an embodiment, the device **12000** may include a heat engine **12016** (e.g., electric coils that produce heat from electricity, FAR infrared heating elements that produce infrared signals) that may output heat from the device **12000** that is used to heat vaginal or skin tissue to be treated prior to and/or during treatment. The tissue treatment device **12000** may further include a user interface **12018** that may be a touch screen, push buttons, knobs, switches, and/or any other device that provides an operator or user with the ability to control functional operation of the device **12000** and components therein. For example, to ensure that the vaginal tissue is properly prepared to be treated by the device

12000, vaginal walls and tissue may be heated for a period of time to ensure that the fascia tissue is prepared to be manipulated during treatment. The heating may be performed by an operator (or user of the device **12000**) turning on a heating element, such as the heat engine **12016**. The user interface **12018** may also include a switch or trigger that controls spin state (e.g., ON state or OFF state) of the motor **12008** along with spin profile of the controller **12011** that controls the motor **12008**. During the pretreatment, the motor **12008** may be in an OFF state, thereby allowing for the effector **12002** to be utilized to perform heating without spin.

(228) Alternative to the heating element **12016** being used to heat the vaginal tissue, the effector **12002** that include a heating mechanism, such as a heating coil **12020** that are encased within a wall **12021** or extend through a cavity defined by the wall **12021** of the effector **12002** to cause an external surface of the wall **12021** of the effector **12002** to heat to a particular temperature (e.g., 108 degrees Fahrenheit). To cause the heating coil **12020** to heat, an electrical connection may be made to the heat engine **12016**, which may control an amount of electrical current to pass through the heating coil **12020**. The electrical connection may be a physical or wireless (e.g., inductor). The wall, which may be formed of a monolithic material, such as stainless steel, silicone, plastic, or combination thereof, may (i) be a single component to avoid seams or other discontinuities to avoid vaginal tissue damage during operation, and (ii) be capable of being sterilized and operate safely in the temperature range (e.g., up to 120 degree Fahrenheit) of the effector. It should be understood that the temperature may be static or the heat engine **12016** may be capable of ramping the temperature up over time (e.g., from 98.6° F. to 110° F.), thereby making the heating treatment easier for the patient. It should be understood that alternative passive (e.g., ceramic) or active (e.g., FAR infrared) heating elements may be configured to be disposed within the effector **12002** so as to apply heat to the patient's vaginal tissue prior to and/or during treatment. It should be understood that heating elements, such as the heating coil **12020**, may also be included in the effectors **12004** and **12006** so that heat may be applied during treatment of the vaginal tissue.

(229) The effector **12002** may be have a smooth outer surface of the wall. Effector **12004** may be configured with protrusions or nubs **12022** along the outside wall. The nubs **12022** may be formed of the same material of a wall **12023** of the effector and be U-shaped, sinusoidal shaped, pyramid shaped (with a curved tip), or have a spline shape to avoid having any sharp edges to minimize risk of damage to vaginal tissue yet perform fascia tissue treatment. In an embodiment, the nubs **12022** may be a different material (e.g., silicone) from that of the wall **12023** of the effector **12022**. In an embodiment, the nubs **12022** may be integrated with a sheath or condom-like cover that is extended over the wall **12023** of the effector, thereby allowing for an effector to be created and different protrusions to be used in a more cost-effective manner. In an embodiment, the sheath may be a stiff or rigid material (e.g., strong plastic or metal) that is attachable to the effector. In the event of the nubs **12022** being pliable (e.g., soft silicone), the effector **12004** may be used for vaginal tissue preparation.

(230) The length of the nubs **12022** (measured perpendicular to the wall of the effector **12002**) may range from a quarter-inch to two-inches depending on the treatment being performed and physical characteristics of the vaginal area of the patient. The diameter of the effector **12002** may range from one-half inch to four-inches. Although shown as a constant diameter along the entirety of the effectors **12002**, **12004**, and **12006**, it should be understood that non-constant diameters and non-linear shaped (e.g., curved with a waist in the middle and bulbous on the distal end from the proximal end that attaches to the shaft **12010**) may also be possible. The length of the effectors **12002**, **12004**, and **12006** may vary in length from 2-inches to 10-inches. Alternative dimensions may also be possible.

(231) The effector **12006** may be configured with tissue treatment elements **12024** connected to or molded with the external surface of a wall **12025** of the effector **12006**. The tissue treatment elements **12024** may have the same or similar configurations as the tissue treatment element of FIGS. **108**A-**108**D, as previously described. In an alternative embodiment, the tissue treatment

elements **12024** may be in the form of fingers that connect directly to the wall **12025** of the effector **12006**. The configuration of the fingers may be similar or the same as those described with regard to FIGS. **113-116**. For example, the fingers may be curved along a shaft of the fingers with a concave surface facing a direction of forward rotation of the effector **12006** while the shaft **12010** is spinning. It should be understood that the motor **12008** may be capable of spinning in both directions. In an embodiment, a vibratory element (not shown) may be disposed with the tissue treatment device **12000** such that the effectors **12004** and **12006** may be capable of vibrating or projecting axially forwards and backwards during treatment of the vaginal tissue. Lubrication of the vaginal tissue may be applied prior to and during treatment to avoid injury to the vaginal tissue.
(232) The principles described herein may include, but is not limited to, features and combinations of features described herein and recited in the following paragraphs. It should be understood that the following paragraphs should not be interpreted as limiting the scope of the claims
(233) One embodiment may include a powered treatment device for treating fascia tissue including a housing, an actuator coupled to and disposed substantially within the housing, the actuator configured to rotate. A tissue treatment assembly may include a panel detachably coupled to the actuator so as to be rotated with respect to the housing when the actuator is in an ON state, and multiple finger members fixedly coupled to the panel at a proximal end of the respective finger members, the finger members being rigid and extending away from the panel, where each one of the finger members may have a respective central axis extending from the proximal end to a distal end of the respective finger member. At least one of the central axes may extend at least partly along a radial direction and/or a circumferential direction of the panel.
(234) The circumferential direction may be oriented along a direction of travel of the panel when the actuator is in the ON state. A portion of each one of the finger members may extend along the circumferential direction. Each of the central axes may extend along the circumferential direction. At least one of the finger members may include a first portion, a second portion, and a third portion, the first portion extending axially away from the panel, the second portion extending away from the first portion at a first angle, and a third portion extending away from the second portion at a second angle and at least partially axially away from the panel.
(235) At least one of the finger members may include a shaft having a convex surface extending between the proximal end and the distal end of the respective finger member, the at least one finger member defining a tip being monolithic with the shaft and at which the convex surface transitions to a concave surface, the concave surface being oriented with a direction of travel of the panel when the actuator is in an ON state.
(236) At least one of the finger members may include a shaft having a convex surface extending between the proximal end and the distal end of the respective elongated finger member, the convex surface facing axially toward the panel. At least one of the finger members may include a concave surface facing axially away from the panel. The panel may be substantially circular and planar, and wherein the finger members define respective tip portions, the tip portions being substantially co-planar and defining a plane that is spaced apart from the panel and substantially perpendicular to a central axis of the panel. In an embodiment, the finger members are elongated.
(237) The powered treatment device may further include a plurality of force sensors configured to measure force applied to quadrants of the panel, and a controller communicably coupled to the force sensors, the controller configured to determine forces being applied to the quadrants of the panel while the actuator is in the ON state based on sensor data from the force sensors.
(238) A user interface may be communicatively coupled to the controller, and be configured to receive signals from the controller to notify a user of the determined forces. The user interface may include at least one light that is configured to visually indicate a difference in forces between two quadrants of the tissue treatment assembly. The actuator may include a direct drive motor that is coupled to the tissue treatment assembly without an intervening gear set.
(239) The powered treatment device may further include a heating device that is configured to

deliver heat toward a side of the tissue treatment assembly when the actuator is in the ON state. The tissue treatment assembly may further include a support member that engages the actuator, wherein the support member is coupled to the panel at a position that is spaced radially apart from a center of the panel. At least one of the finger members may extend along a first circumferential direction relative to a central axis of the panel, the actuator configured to rotate the panel along the first circumferential direction.

(240) A tissue treatment assembly may include a panel, a support member disposed on the panel, where the support member may include a connecting element configured to detachably couple the support member to an actuator. Multiple finger members may be coupled to the panel at a proximal end of the finger members, where the finger members may be rigid and extend away from the panel. Each one of the plurality of finger members may have a respective central axis extending from the proximal end to a distal end of the respective finger member. At least one of the central axes may extend at least partly along a radial direction and/or a circumferential direction of the panel.

(241) The finger members may define respective tip portions, where the tip portions may be substantially co-planar and disposed along a reference plane that is spaced apart from the panel and that is substantially perpendicular to a central axis of the panel. A size of a first one of the tip portions may be different from a size of a second one of the tip portions. The panel may be substantially planar. The finger members may be arranged in approximately equal intervals along a surface of the panel.

(242) At least one of the finger members may extend radially beyond an outer perimeter of the panel. Each of the central axes may extend along the circumferential direction. At least one of the finger members may include a first portion extending axially away from the panel, a second portion extending at an angle relative to the first portion and substantially parallel to the panel, and a third portion extending at least partially axially away from the second portion.

(243) The panel may define an opening at a location of at least one of the finger members. At least one of the finger members may include a shaft having a convex surface extending between the proximal end and the distal end of the respective finger member, the finger member(s) defining a tip being monolithic with the shaft and at which the convex surface transitions to a concave surface, the concave surface being orientated with a circumferential direction of the panel.

(244) The finger members may be arranged in multiple circumferential rows along the panel. Adjacent rows of the finger members may be rotationally offset from one another. The finger members may be arranged in a spiral pattern along the panel. At least one of the finger members may curve back toward the panel so that the respective finger member engages the panel at both the proximal end and the distal end of the respective finger member. At least one of the finger members may include a first edge extending substantially axially away from the panel and a second edge opposite the first edge that curves back toward the panel.

(245) The finger members may form part of a tissue treatment element that is detachably coupled to the panel. The finger members may be integrally formed with the panel as a single monolithic piece. At least one of the finger members may define a frustoconical shape. The support member may be coupled to the panel at a position that is spaced radially apart from a center of the panel. The support member may further include multiple cylindrical support ribs extending at least partially axially away from the panel, the cylindrical support ribs may be concentric with a central axis of the panel. The support member may further include at least one substantially planar rib that extends radially between adjacent ones of the cylindrical support ribs. The support member may further include a skeletal framework that is configured to distribute loading across the panel.

(246) The connecting element may be at least partially disposed at one of: (i) an outer perimeter of the panel, or (ii) an intermediate radial position between a center of the panel and the outer perimeter of the panel. The panel may be curved along a radial direction in a neutral position so that a center of the panel protrudes axially away from an outer perimeter of the panel. The panel

may be made from a flexible material, wherein the center of the panel is axially alignable with the outer perimeter of the panel in response to an axial force applied to the center of the panel.
(247) The previous description is of a preferred embodiment for implementing the invention, and the scope of the invention should not necessarily be limited by this description. The scope of the present invention is instead defined by the following claims.

Claims

1. A powered treatment device for treating fascia tissue, comprising: a housing; an actuator coupled to and disposed substantially within the housing, the actuator configured to rotate; a tissue treatment assembly, including: a panel detachably coupled to the actuator so as to be rotated with respect to the housing when the actuator is in an ON state; and a plurality of finger members fixedly coupled to the panel at a proximal end of the respective finger members, the finger members being rigid and extending away from the panel, each one of the plurality of finger members having a respective central axis extending from the proximal end to a distal end of the respective finger member away from the panel, the central axes of the finger members extending at least partly along a common circumferential direction of the panel.
2. The powered treatment device according to claim 1, wherein the circumferential direction is oriented along a direction of travel of the panel when the actuator is in the ON state.
3. The powered treatment device according to claim 1, wherein the finger members include: a first portion of extending parallel to the panel; and a second portion extending from a distal end of the first portion and axially away from the panel.
4. The powered treatment device according to claim 1, wherein at least one of the finger members includes a first portion, a second portion, and a third portion, the first portion extending axially away from the panel, the second portion extending away from the first portion at a first angle, and the third portion extending away from the second portion at a second angle and at least partially axially away from the panel.
5. The powered treatment device according to claim 1, wherein at least one of the finger members includes a shaft having a convex surface extending between the proximal end and the distal end of the respective finger member, the at least one finger member defining a tip being monolithic with the shaft and at which the convex surface transitions to a concave surface.
6. The powered treatment device of claim 5, wherein the concave surface is oriented with a direction of travel of the panel when the actuator is in an ON state.
7. The powered treatment device according to claim 1, wherein at least one of the finger members includes a shaft having a convex surface extending between the proximal end and the distal end of the respective elongated finger member, the convex surface facing axially toward the panel.
8. The powered treatment device according to claim 1, wherein at least one of the finger members includes a concave surface facing axially away from the panel.
9. The powered treatment device according to claim 1, wherein the panel is substantially circular and planar, and wherein the finger members define respective tip portions, the tip portions being substantially co-planar and defining a plane that is spaced apart from the panel and substantially perpendicular to a central axis of the panel.
10. The powered treatment device according to claim 1, wherein the finger members are elongated.
11. The powered treatment device according to claim 1, further comprising: a plurality of force sensors configured to measure force applied to quadrants of the panel; and a controller communicably coupled to the force sensors, the controller configured to determine forces being applied to the quadrants of the panel while the actuator is in the ON state based on sensor data from the force sensors.
12. The powered treatment device according to claim 11, further comprising a user interface communicatively coupled to the controller, and configured to receive signals from the controller to

notify a user of the determined forces.
13. The powered treatment device according to claim 12, wherein the user interface comprises at least one light that is configured to visually indicate a difference in forces between two quadrants of the tissue treatment assembly.
14. The powered treatment device according to claim 1, wherein the actuator includes a direct drive motor that is coupled to the tissue treatment assembly without an intervening gear set.
15. The powered treatment device according to claim 1, further comprising a heating device that is configured to deliver heat toward a side of the tissue treatment assembly when the actuator is in the ON state.
16. The powered treatment device according to claim 1, wherein at least one of the finger members extends along a first circumferential direction relative to a central axis of the panel, the actuator configured to rotate the panel along the first circumferential direction.
17. The powered treatment device of claim 1, wherein a height of the finger members normal to the panel is greater than a dimension of the finger members engaged with the panel.
18. A tissue treatment assembly, comprising: a panel; a support member disposed on the panel, the support member including a connecting element configured to detachably couple the support member to an actuator; and a plurality of finger members coupled to the panel at a proximal end of the respective finger members, the finger members being rigid and extending away from the panel, each one of the plurality of finger members having a respective central axis extending from the proximal end to a distal end of the respective finger member away from the panel, the central axes of the finger members extending at least partly along a common circumferential direction of the panel.
19. The tissue treatment assembly according to claim 18, wherein the finger members define a plurality of tip portions, the tip portions being substantially co-planar and disposed along a reference plane that is spaced apart from the panel and that is substantially perpendicular to a central axis of the panel.
20. The tissue treatment assembly according to claim 18, wherein the panel is substantially planar.
21. The tissue treatment assembly according to claim 18, wherein the finger members are arranged in approximately equal intervals along a surface of the panel.
22. The tissue treatment assembly according to claim 18, wherein at least one of the finger members includes a first portion extending axially away from the panel, a second portion extending at an angle relative to the first portion and substantially parallel to the panel, and a third portion extending at least partially axially away from the second portion.
23. The tissue treatment assembly according to claim 18, wherein the panel defines an opening at a location of at least one of the finger members.
24. The tissue treatment assembly according to claim 18, wherein at least one of the finger members includes a shaft having a convex surface extending between the proximal end and the distal end of the respective finger member, the at least one finger member defining a tip being monolithic with the shaft and at which the convex surface transitions to a concave surface, the concave surface being orientated with the circumferential direction.
25. The tissue treatment assembly according to claim 18, wherein the finger members are arranged in a plurality of circumferential rows along the panel.
26. The tissue treatment assembly according to claim 25, wherein adjacent rows of the finger members are rotationally offset from one another.
27. The tissue treatment assembly according to claim 18, wherein the finger members form part of a tissue treatment element that is detachably coupled to the panel.
28. The tissue treatment assembly according to claim 18, wherein the finger members are integrally formed with the panel as a single monolithic piece.